

TfNSW Bridge Asset Management Assurance and Prioritisation Final report

FINAL

May 2026

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Acknowledgement of Country

Transport pays respect to Elders past and present, and recognises and celebrates the diversity of Aboriginal peoples and their ongoing cultures and connections to the lands and waters of NSW.

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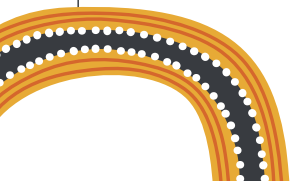
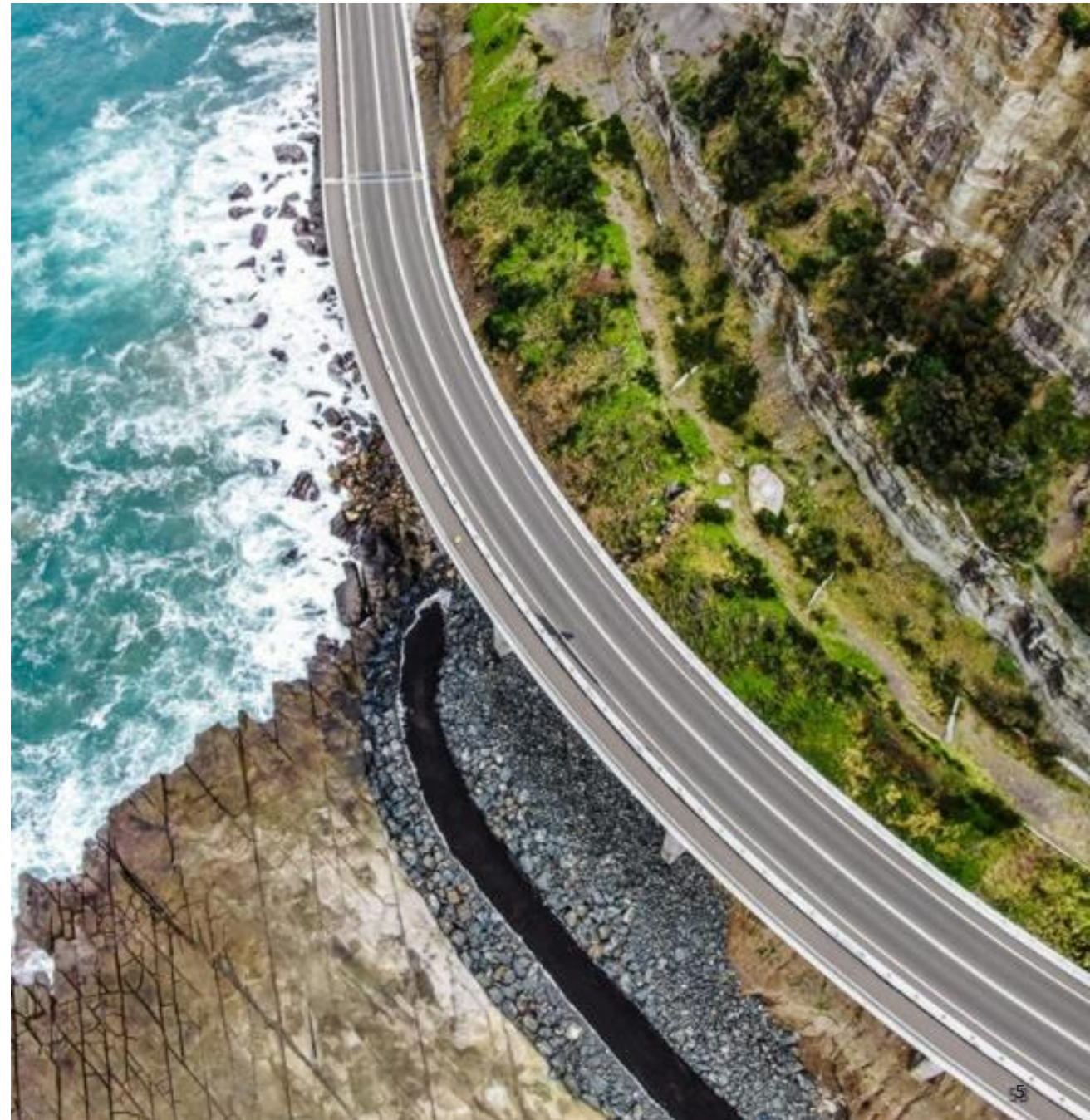
01

Project Context and Purpose



Current state

- Bridge data is not well defined and centralised to enable prioritisation across all modes
- Current ASPs do not contain the level of detail required for prioritisation assessment
- Ad hoc funding requests for bridge maintenance work appear outside of the ASP process
- Bridge condition has not been audited or validated or updated consistently across all bridge assets
- Bridge prioritisation focused solely on asset life and do not always consider customer and network impacts
- Governance and decision-making is disparate e.g. limiting loads on bridges



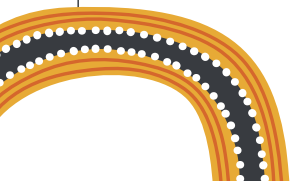
Project objective and value to TfNSW

Objective:

Create a defensible and transparent prioritisation framework that can be applied to all transport bridges that is reflective of asset performance, risk and customers needs that is aligned to industry-accepted methods

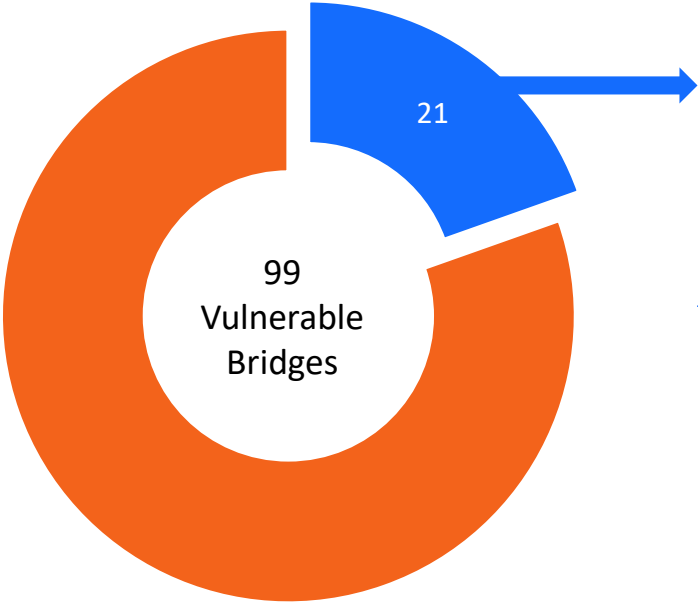
Value to TfNSW:

- Customer focused bridge prioritisation rather than focus on asset condition
- Centralised and consistent prioritisation of bridges across all modes
- Identify and address bridge data requirements and quality challenges (to inform data standards and TACP design)
- Provide evidence base to inform funding requests for bridge asset lifecycle investment needs
- Prioritisation framework is scalable to all asset classes to strengthen quality of ASPs to demonstrate alignment to transport service outcomes



Background: Initial Assessment

- In 2024 the Regional Investment team commenced an initial assessment and identified 99 vulnerable bridges based on input provided by RMR and PAM.
- Of the 99 vulnerable bridges, 21 were evaluated as priority based on a high-level assessment using the draft Strategic Development Review completed by PAM.
- It was acknowledged that this initial assessment needs to be expanded to consider the entire Transport bridge portfolio with a customer focused prioritisation tool.

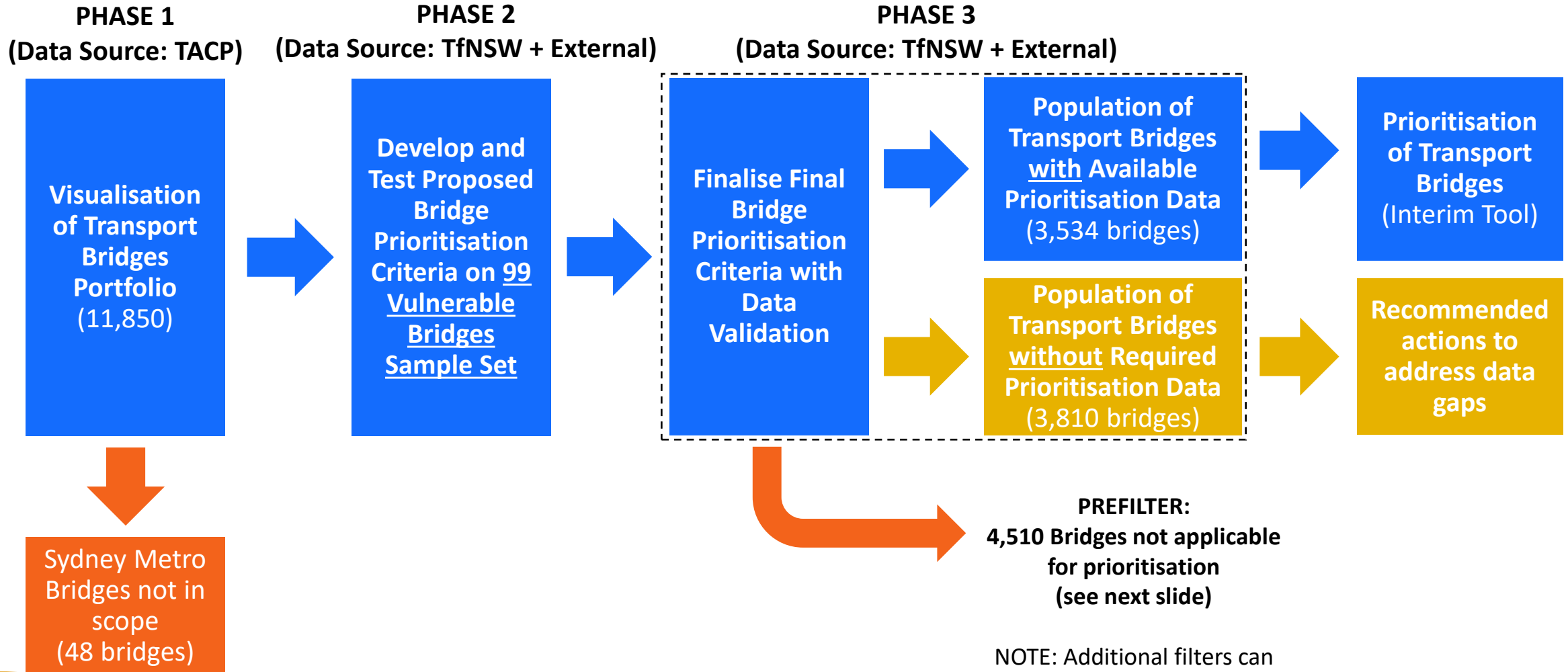


Greater Sydney	
B98	Bentley's Bridge, Elizabeth Bay
B44	Pacific Hwy, Turramurra OBGE
B12555	Meadowbank footbridge
TOTAL = 3 out of 43	
Regional South	
B938	Springhill Road, Port Kembla
B941	JJ Mathews bridges, Wollongong
B942	JJ Mathews bridges, Wollongong
B3244	Tooleybuc Bridge
B6298	Sheehan Bridge, Hume Hwy
B3215	Swan Hill
B875	Hampden Bridge
TOTAL = 7 out of 16	
Regional West	
B972	Great Western Hwy, Linden
B4122	Angus Bridge, Mitchell Hwy
CRN	Lachlan Valley Way, Yass Junction
B4767	Boothaguy Creek Bridge
B4660	Scabbing Flat Bridge
TOTAL = 5 out of 10	
Regional North	
B1365	Swansea northbound
CRN	Industrial Drive, Port Waratah
CRN	Maud Street, Sandgate
CRN	Church Street, Maitland
B1518	Forster / Tuncurry
B2322	Old Grafton bridge
TOTAL = 6 out of 30	

Approach and Milestones

	Phase 0: Initial Assessment of Vulnerable Bridges COMPLETED MARCH 2025	Phase 1: Provide visibility of all Transport bridges COMPLETED SEPTEMBER 2025	Phase 2: Evaluate current bridge data and prioritisation methods COMPLETED FEBRUARY 2026	Phase 3: Develop customer-focused bridge prioritisation COMPLETED MAY 2026
Key Activities	- First pass assessment on vulnerable road bridges identified by RMR and PAM - Initial prioritisation assessment considering the Strategic Development Review completed by PAM - Assessment did not have information on the current condition issues and excluded rail bridges	- Data collation from TACP for all Transport bridges - Develop bridge visualisation dashboard showing Transport bridge population by location across NSW - Provide visibility of non-Transport owned structures that may impact Transport services on bridges	- Review existing Transport bridge prioritisation methodologies - Co-develop customer focused bridge prioritisation criteria and methodology with stakeholders - Gap analysis on bridge prioritisation data requirements - Develop interim Prioritised Bridge List (based on 99 vulnerable bridges from initial assessment in Phase 0)	- Bridge prioritisation data validation and mapping across TfNSW and external sources - Finalise customer focused bridge prioritisation methodology and tool - Assess priority bridges from Transport bridge population based on developed methodology/tool - Final report containing findings and recommendations
Deliverables	99 vulnerable bridges identified (RMR and PAM) 21 preliminary priority bridges identified (previous page)	Visualisation dashboard (using TfNSW PowerBI platform) of all Transport bridges (refer to Appendix)	- Interim Bridge Prioritisation List (refer to Appendix) - Data Gap Analysis (refer to Section 3)	- Interim Bridge Prioritisation Tool (refer to Section 4) - Prioritised list of Transport bridge portfolio (refer to Section 4)

Bridge prioritisation process



Pre-filter

- The pre-filter enables TfNSW to exclude bridges from the prioritisation framework that are not applicable to inform investment decisions.
- The pre-filter criteria shown was developed with TfNSW stakeholders (refer to slide 11 for further details) to inform the design of the prioritisation framework.
- The pre-filter criteria is applied to the population of bridges in TACP, prior to prioritisation.

BRIDGES EXCLUDED FROM PRIORITISATION:



Not under TfNSW's responsibility for maintenance, repair, or investment



Already approved for renewal or replacement to meet current standards



Recently upgraded, having undergone major capital investment within the past year and already meeting long-term service needs



Fauna crossings



Bridges less than 50 years old are excluded from the interim prioritisation¹

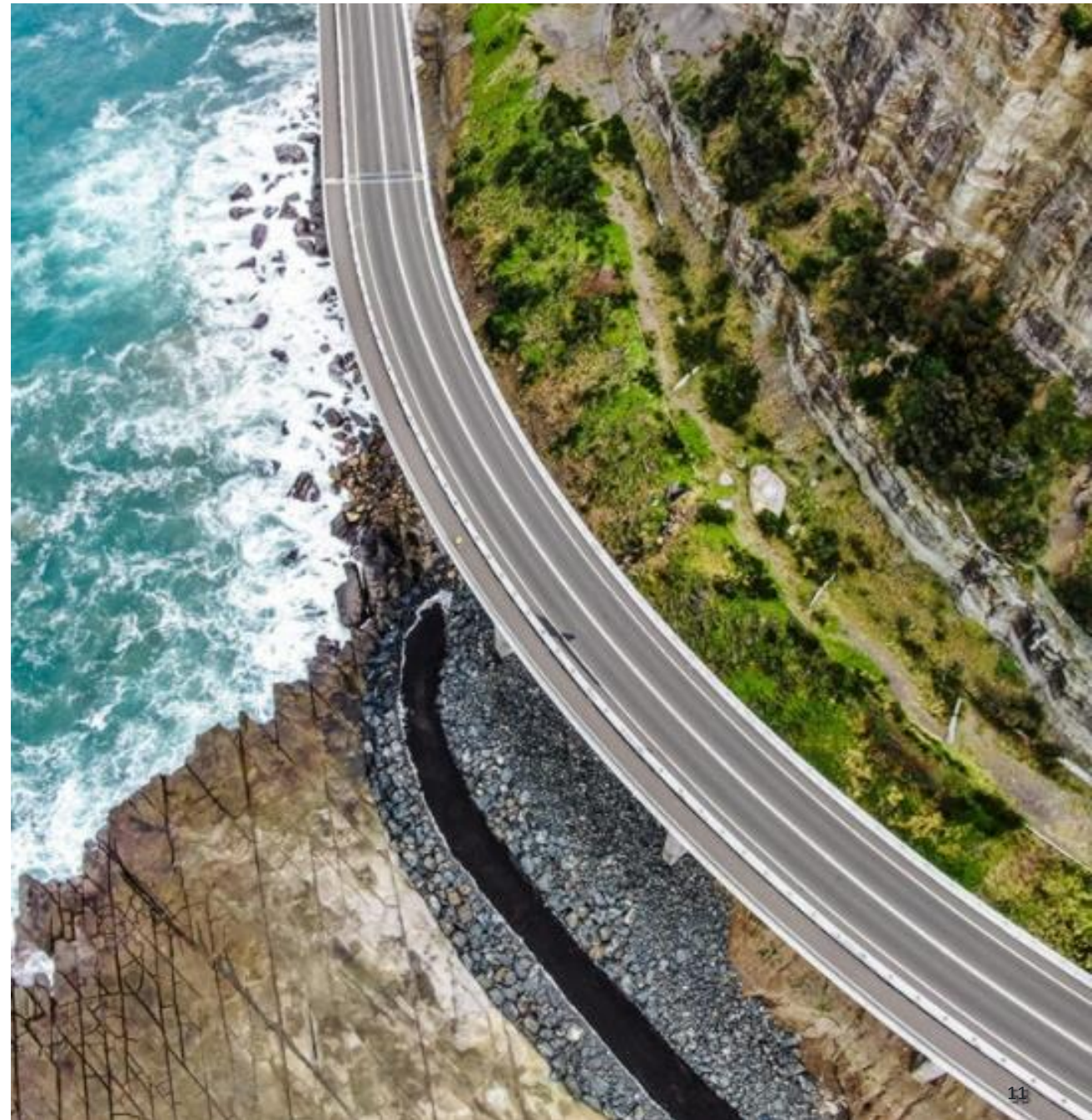
1. This criterion can be adjusted based any value. 50 years was the value requested by TfNSW Prioritisation and Asset Management Branch to align with known issues with engineering standards for older bridges.

Stakeholder Engagement

The bridge prioritisation methodology and been developed with engagement and input from Transport business areas, including:

- Prioritisation and Asset Management
- Road Maintenance and Resilience
- Sydney Trains
- Active Transport and Vibrancy
- Freight
- Regional Integration

NOTE: Bridge prioritisation analysis has been limited due to Sydney Trains being unable to provide rail bridge prioritisation data for the engagement to date.

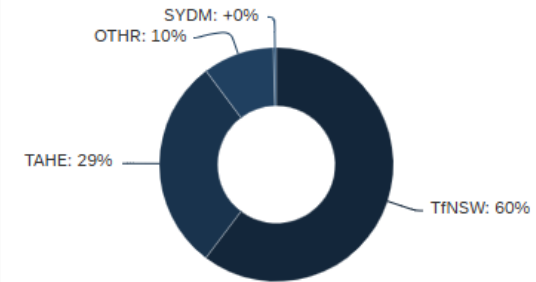


Portfolio overview

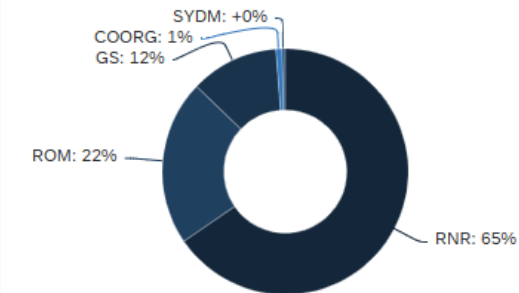
Current portfolio: 11,850 bridges

- Bridge portfolio identified using the Transport Asset Custodian Platform (TACP) centralised asset register
- TACP bridges asset data sourced from 13 maintainer systems across multiple Transport modes
- Condition data is limited and not validated. TACP Tranche 2 design and implementation will consolidate additional asset condition data to support prioritisation
- Bridge prioritisation will identify additional asset attributes to inform improved decision making and support TACP Tranche 2 design.
- The following bridges are not in scope:
 - Sydney Metro
 - Fauna crossings classified as bridges

Network Assets by Asset Owner

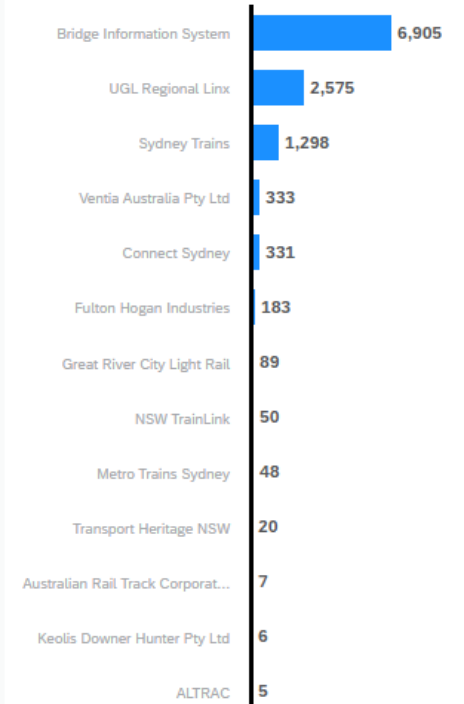


Network Assets by Asset Custodian



Note: Asset Owner and Custodian data is currently undergoing updates in line with the organisational structure

Number of Network Assets by Maintainer/EAM



*Bridge Information System contains RMR internally maintained bridge data

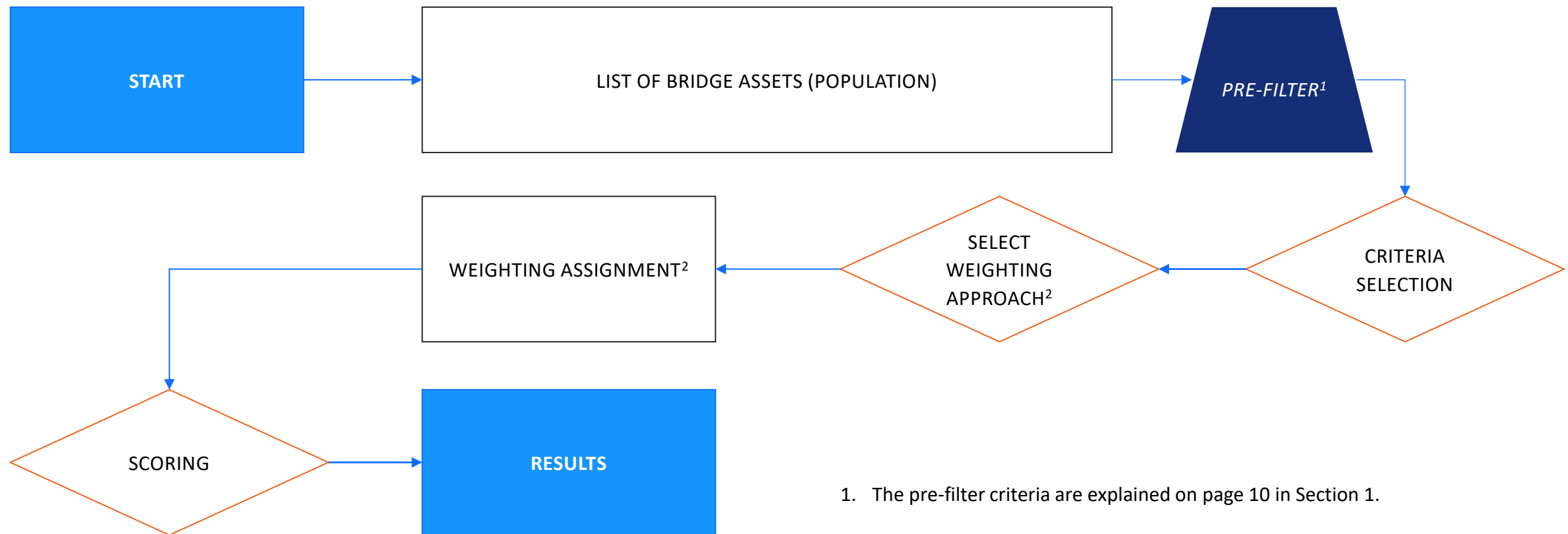
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The Proposed Prioritisation Framework



The proposed prioritisation framework is based on a Multi-Criteria Analysis approach

“Multi-criteria analysis (MCA) is a tool that can be used to compare reform and investment proposals” -- Infrastructure Australia

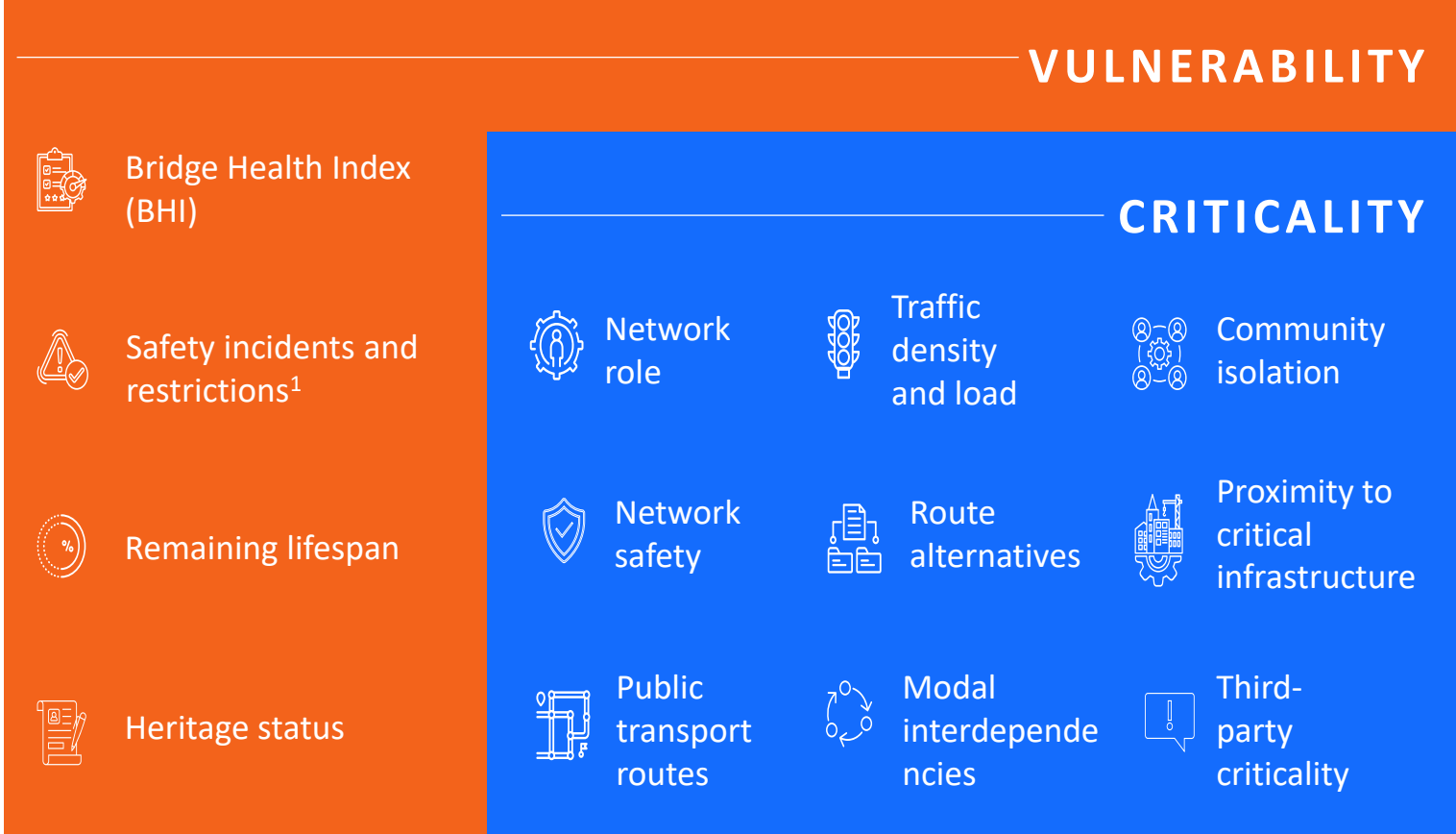


1. The pre-filter criteria are explained on page 10 in Section 1.

2. Weightings have been applied as part of the interim prioritisation application of the current framework (see Section 4.a), and are available as an option for TfNSW to amend in future iterations

Proposed bridge prioritisation criteria summary

Multi-criteria analysis (MCA) tool was developed for the proposed bridge prioritisation based on the criteria shown below. Refer to page 17 and 18 for definitions of each criteria.



CRITICALITY – *asset function*
Customer and service outcomes, enablement

VULNERABILITY – *asset condition*
Asset status, health and condition measures

1. This criterion was renamed and its definition amended following feedback on the Pilot Phase outputs. The previous criterion “Historical safety incidents” is included in the Appendix material.

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Customer user types for the criticality criteria

Development of bridge 'customer user types' approach for criticality

For each of the **Criticality** criteria¹, the following bridge user types will be evaluated:

- Road transport – passenger vehicles and light commercial vehicles
- Road transport – Class 2 and Class 3 heavy vehicles (NHVR)
- Road transport – Class 1 heavy vehicles (NHVR)
- Rail transport – passenger trains
- Rail transport – freight trains
- Active transport – pedestrians
- Active transport – cyclists
- Water transport – commuter ferries and private vessels
- Water transport – freight shipping and other commercial vessels

1. *Community isolation, Proximity to critical infrastructure and Third-party criticality do not vary by user type; these are applicable to the full bridge, and should be evaluated relative to 'Road transport – passenger vehicles and light commercial vehicles' as a reference point*

Proposed criteria and definitions

CRITICALITY – Customer and service outcomes, enablement

CRITICALITY	
Network role	Importance of the bridge in the broader transport network. Reflects the role of the bridge in maintaining national, state and local transport continuity. Includes if the bridge forms part of the National Land Transport Network, a critical freight or passenger corridor, or a Strategic Cycleway Corridor.
Network safety	The degree to which the bridge supports network safety outcomes, including emergency management and evacuation functions. Considers contribution of the bridge to safety by design hierarchy (e.g., avoid level crossings, grade or modal separation).
Public transport routes	The bridge's contribution to public transport continuity, including available seat kilometres, school bus routes, and intermodal connectivity.
Traffic density and load	Loading (damage) of traffic using the bridge and traffic volume of passenger users.
Route alternatives	Availability of viable alternative routes or crossings if the bridge becomes unavailable, and the delay imposed (detour time).
Modal interdependencies	Assesses how the performance or failure of a bridge may contribute to the mode shift of users. Modal shift will result in broader network impacts, including reliability or delay in the short-term, and increased asset degradation rates in the long-term.
Proximity to critical services	The bridge's role in maintaining access to essential and emergency services, including hospitals, schools. Considers impact on community resilience and continuity of supply during disruptions.
Third-party criticality	Quantum and importance of third-party services (e.g., utilities, pipelines, communication assets) or other structures (e.g., carparks, runways) that the bridge supports, where bridge failure would have broader system impacts.
Community isolation	The population that is isolated if a bridge fails, beyond just route alternatives.

NOTE: Refer to Appendix 4.b for the detailed criteria hurdle rate definitions developed as examples prior to the Pilot Phase

Proposed criteria and definitions

VULNERABILITY – Asset status, health and condition measures

VULNERABILITY	
Bridge Health Index (BHI)	A broad qualitative indicator of potential risk of structural failure derived from the condition ratings of the bridge elements from the latest inspection of a bridge
Safety incidents and restrictions¹	Measure of severity, frequency and outcome of recorded recent incidents resulting in bridge restrictions (such as load/dimension limits) caused by bridge condition/defects or safety-related issues
Remaining lifespan	Remaining life of the bridge relative to the total asset design life, expressed as a proportion, accounting for any major maintenance/replacement activities conducted
Heritage status	Is the bridge heritage listed? At what level is it heritage listed?

1. This criterion was renamed and its definition amended following feedback on the Pilot Phase outputs. The previous criterion “Historical safety incidents” is included in the Appendix material.

NOTE: Refer to Appendix 4.b for the detailed criteria hurdle rate definitions developed as examples prior to the Pilot Phase

Load or Dimension Limited Bridges

User types allow for nuanced specificity and temporal-variation in bridge access permissions and limitations

- Load and dimension access limits (including updates to those limits following events or interventions) are captured through an attribute in the base data set for use by the prioritisation framework where data is available.
- Access limits imposed on bridges restricting access to customers will impact the **safety incidents and restrictions** scores based on the recorded recent incidents resulting in bridge restrictions (such as load/dimension limits) caused by bridge condition/defects safety related issues.
- In addition, access restrictions which restrict user types across the bridge will impact the **route alternative** scores applied for the user type.
 - For example, this would be reflected in PBS vehicle access maps; a bridge that becomes inaccessible would be removed from the permitted routes map, significantly impacting travel time measure through the route alternative criteria.
 - This currently only applies to road freight user types and not rail freight due to complexity of modelling route alternatives for freight rail customers.

03

CURRENT DATA QUALITY CHALLENGES



Data Quality Assessment (based on TACP data)

TACP identifies **11,850 bridges** that TfNSW organisations are responsible for. The information in the TACP databases has come from the numerous sources. Data gaps in TACP indicate missing data from the external original collated source information.

Road and Rail bridges of various types (steel, timber, brick etc) make up the majority of the portfolio across the state. Other bridges in the portfolio include:

- pedestrian foot bridges and subway underpasses,
- stock underpasses and other fauna passageways,
- bridges that support services (water, telco and power), and
- service crossing on industrial lands.

Multiple maintainers have well embedded, approved processes and methodologies for assessing condition of these assets. The reported measure of condition is inconsistent: some maintainers use a Bridge Health Index (BHI) other do not.

Bridge locations have multiple transport types at the same site. Bridges can have road, rail, pedestrian surface or subway paths etc structural elements. Each asset type is recorded separately not indicating a parent child relationship.

All of the customer attribute data inputs into the Bridge Prioritisation Framework do not currently reside in the TACP database.

11,850 bridges

that TfNSW organisations are responsible for

94%

have the **Asset Owner** identified, 6% classified as “Others”

100%

have the **Asset Custodian** identified

42%

have the **Primary Maintainer** identified, 58% associated with Road Network Resilience

<70%

have the **Construction Date** or **Commission Date** fields completed. As a bridge may be beyond its original design life, upgrades, renewal and appropriate maintenance can extend operating life.

88%

have **GIS coordinates** completed

0%

have the date of the **Last Condition Assessment** identified in the TACP dataset. This information will need to be updated across the portfolio on a 6-12 monthly basis

42%

have **Asset Criticality** identified as “Null”. Criticality is described and used differently by different operators and maintainers. It does not provide consistent application as it references road traffic ranking, road class description, number of timetabled services

74%

have the specific **Location** field completed. Associated information (e.g., GIS coordinates, location corridor, asset description including a location) minimises the location error gap

22%

have a **Condition** identified as “Null”. This information will need to be updated across the portfolio on a 6-12 monthly basis

Data availability and confidence by user type and criterion – *Interim Prioritisation*

The prioritisation criteria were assessed for application based on the availability and quality of data by TfNSW and relevant external sources. Results below provide the data quality availability and confidence rating of each criteria. Refer to next page for further details.

User Type	Bridge Health Index (BHI)	Safety incident and restrictions	Remaining lifespan	Heritage status	Network role	Network safety	Public transport routes	Traffic density and load	Route alternatives	Modal interdependencies	Proximity to critical infrastructure	Third-party criticality	Community isolation
Over – Road – Passenger	Medium	Unavailable	Low	Unavailable	High	Unavailable	Medium	Low	Medium	Unavailable	Unavailable	Unavailable	Unavailable
Over – Road – Heavy vehicles Class 2 & 3					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Over – Road – Heavy vehicles Class 1					Unavailable	Unavailable	N/A	Unavailable	Unavailable	Unavailable	N/A		N/A
Over – Active Transport – Pedestrians					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Over – Active Transport – Cyclists					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Over – Rail – Passenger trains					Unavailable	Unavailable	High	Unavailable	Unavailable	Unavailable	N/A		N/A
Over – Rail – Freight trains					Unavailable	Unavailable	N/A	Unavailable	Unavailable	Unavailable	N/A		N/A
Under – Road – Passenger					Medium	Unavailable	Medium	Low	Medium	Unavailable	Unavailable		Unavailable
Under – Road – Heavy vehicles Class 2 & 3					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Under – Road – Heavy vehicles Class 1					Unavailable	Unavailable	N/A	Unavailable	Unavailable	Unavailable	N/A		N/A
Under – Active Transport – Pedestrians					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Under - Active Transport – Cyclists					Unavailable	Unavailable	N/A	Unavailable	Medium	Unavailable	N/A		N/A
Under - Rail - Passenger trains					Unavailable	Unavailable	High	Unavailable	Unavailable	Unavailable	N/A		N/A
Under - Rail - Freight trains					Unavailable	Unavailable	N/A	Unavailable	Unavailable	Unavailable	N/A		N/A
Under- Water Transport – Passenger & private vessels					Unavailable	Unavailable	Unavailable	Unavailable	Unavailable	Unavailable	N/A		N/A
Under - Water Transport – Freight & commercial vessels					Unavailable	Unavailable	N/A	Unavailable	Unavailable	Unavailable	N/A		N/A

Note: **Third-party criticality** does not vary by user type. While it is a **Criticality** criterion (i.e., it relates to the function of the bridge, not the physical attributes of the bridge), the bridge is evaluated on the whole for this criterion, similar to the four **Vulnerability** criteria. **Community isolation** and **Proximity to critical infrastructure** are only evaluated for *Road – Passenger* user types (both Over and Under the bridge); other user types are not relevant for these criteria. There is some variation between Over and Under for user types in **Network Role** due to varying data availability and confidence.

Data availability and confidence summary by criterion – *Interim Prioritisation*

A summary of the assessed data quality for the interim bridge prioritisation data inputs, including identified limitations, is presented below. Recommendations in response to the findings outlined below are consolidated in Section 5.

Criterion	Data Availability & Confidence	Findings / Limitations
Bridge Health Index (BHI)	Medium	Condition data for road bridges was available for ~2400 bridges and could be mapped directly and automatically to vulnerability scores, with gaps clearly identifiable where data was missing. Refer to page 21 for further details on TACP data quality.
Safety incidents & restrictions	Unavailable	Safety incident data was not available for this analysis and was not applied to the interim prioritised list. Current access restrictions and closure (such as load and dimensional limits) data was not available in a structured and up to date form that captured severity, duration and recurrence. This criterion has therefore been excluded from the interim prioritisation.
Remaining lifespan	Low	Age-based proxies from BIS and CRNOB enabled broad identification of remaining life for the pilot phase, albeit with varying confidence where design life data was incomplete. Given the varying confidence in the design life data, the interim prioritisation instead applied a filter-based approach at the outset, only progressing bridges greater than 50 years old to the detailed prioritisation, or otherwise those bridges previously included in the pilot phase. Refer to page 21 for further details on TACP data quality.
Heritage status	Unavailable	National, state and local heritage registers provided a largely-reliable basis for scoring during the pilot phase, with only minor gaps requiring clarification of source-of-truth. The disparate, dispersed and inconsistent nature of the registers and data format (e.g., text-based lists, data tables, geospatial envelopes requiring manual interpretation) limited the ability to automate the evaluation of this criterion across a large bridge portfolio during the interim prioritisation. This criterion has therefore been excluded from the interim prioritisation.

Data availability and confidence summary by criterion – *Interim Prioritisation*

A summary of the assessed data quality for the interim bridge prioritisation data inputs, including identified limitations, is presented below. Recommendations in response to the findings outlined below are consolidated in Section 5.

Criterion	Data Availability & Confidence	Findings / Limitations
Network role	Medium	Application of this criterion for bridges with 'Road – Passenger' users was largely achievable for 'over' users, as road hierarchical and functional classifications were available and therefore scoring could be automated. However, the road classification of a road under a bridge was less reliable, as the road itself was not always readily identifiable. Other user types, including rail mode users and active transport users, did not have complete or available data sets available; during the pilot phase these user type functional classifications were estimated by visual inspection, however this approach was unable to be automated for the interim prioritisation. This criterion has therefore been excluded from the interim prioritisation for user types other than 'Road – Passenger' users.
Network safety	Unavailable	While highly relevant, this criterion was difficult to quantify objectively across all modes and exhibited strong overlap with other criteria. The need to identify a counterfactual safety outcome by user type for each bridge as well as the current safety outcome is difficult to automate using available data sets. This criterion has therefore been excluded from the interim prioritisation.
Public transport routes	Medium	Transport network maps and timetables were sourced to estimate the number of public transport services over or under each bridge for both 'Road – Passenger' and 'Rail – Passenger' user types.
Traffic density and load	Low	'Road – Passenger' traffic volume forecasts were taken from Transport's Regional Travel Model (RTM) for the regional bridges, and AADT values for BIS bridges otherwise applied where available. The accuracy or currency of the AADT data was unable to be confirmed, nor was the data complete. Granular <u>rail movement data</u> is available to TfNSW, however due to its commercially-sensitive nature, this was not available for inclusion in either the pilot or the interim prioritisation; no rail user types were scored on this criterion . <u>Active transport user volumes</u> were estimated visually from spatial context (i.e., adjacent facilities, nature of active transport infrastructure present) for the pilot phase. However, no deterministic dataset was available for automated evaluation in the interim prioritisation's generation; this criterion was not included for active transport user types in the interim prioritisation.

Data availability and confidence summary by criterion – *Interim Prioritisation*

A summary of the assessed data quality for the interim bridge prioritisation data inputs, including identified limitations, is presented below. Recommendations in response to the findings outlined below are consolidated in Section 5.

Criterion	Data Availability & Confidence	Findings / Limitations
Route alternatives	Medium	For the pilot phase, detour times could be estimated using mapping tools, but results were sensitive to time-of-day, direction of travel and vehicle class, requiring manual interpretation. Automation processes applied during the interim prioritisation were generally unable to account for local nuance or circumstance; bridge “start” or “end” points may have obscured appropriate or available detour paths, therefore over- or understating the true detour time. For regional bridges, the ‘Road – Passenger’ detour times were taken from Transport’s established regional detour modelling. This improved the alignment of the route alternative scores to existing Transport datasets. The methodology for this regional detour modelling assumes trips are taken from ‘activity centre’ to ‘activity centre’, but without using the bridge in question; the automated metropolitan approximations adopted for the non-regional bridges instead required the assumption that trips were from one end of the bridge structure to the other, without utilising the bridge. The approximation of ‘Road – Heavy vehicles Class 2 & 3’ and both ‘Active transport – [...]’ user types’ detour times were automated across the state (both regional and metropolitan) using the same methodology as applied for the metropolitan ‘Road – Passenger’ detour calculations. For the ‘Road – Heavy vehicles Class 2 & 3’ user type, detours were calculated for two representative PBS vehicles, and the average taken. Rail detour availability has additional complexity – an alternate route may be technically feasible, but operationally impossible – and therefore this user type was not evaluated in the interim prioritisation.
Modal interdependencies	Unavailable	No reliable, scalable datasets exist to quantify mode shift impacts consistently across the network. Furthermore, mode shift behaviours are largely location specific, and temporally influenced and variate. This criterion has therefore been excluded from the interim prioritisation

Data availability and confidence summary by criterion – *Interim Prioritisation*

A summary of the assessed data quality for the interim bridge prioritisation data inputs, including identified limitations, is presented below. Recommendations in response to the findings outlined below are consolidated in Section 5.

Criterion	Data Availability & Confidence	Findings / Limitations
Proximity to critical services	Unavailable	Spatial proximity could be estimated during the pilot phase, but interaction effects with route alternatives and service criticality required judgement rather than automated scoring. This criterion has therefore been excluded from the interim prioritisation
Third-party criticality	Unavailable	Identification of utilities and services relied heavily on visual inspection during the pilot phase. Limited consistent and exhaustive registers are available for automation, with low confidence in assessing relative importance and consequence of assets. This criterion has therefore been excluded from the interim prioritisation
Community isolation	Unavailable	Applicable only to a small subset of bridges, with most assets scoring zero or low in the pilot phase, raising questions about proportionality versus effort. Limited geospatial datasets mapping the population isolated by a bridge's (Road – Passenger) inaccessibility exist. This criterion has therefore been excluded from the interim prioritisation

Data availability and confidence summary by user type – *Interim Prioritisation*

A summary of the status for the interim bridge prioritisation data inputs, including identified limitations, is presented below.

Limited freight and rail data availability

- Besides an approximation of the detour required for 'Road – Heavy vehicle Class 2 & 3' users, **no freight-related user types** are scored in the interim prioritisation, across any criteria
- While 'Rail – Passenger' public transport routes have been estimated with reasonable confidence, all other **rail user types have been omitted** from the interim prioritisation
- Interim prioritisation results are therefore dominated by bridges with a 'Road – Passenger' user type present; **rail-only or freight-only bridges are not prioritised due to data unavailability**

Road freight user types non-representative

- Ongoing engagement with TfNSW freight stakeholders and review of available datasets have indicated that the adopted generalisation of road freight users into two broad user types (Class 2 & 3 and Class 1 heavy vehicles) does not reflect access and usage of the NSW bridge network, nor do these categorisations match the NHVR access maps.

Recommendations in response to the findings outlined here are consolidated in Section 5

Implications for prioritisation outcomes

The identified data limitations influenced how prioritisation results should be interpreted and applied

Impact on prioritisation results

- Some criteria could only be applied to subsets of bridges, meaning results reflect best-available evidence rather than a fully populated dataset.
- Number of data quality challenges and gaps resulted in a number of bridges being omitted from the results, focusing on prioritisation of bridges where data confidence is high.
- Data resolution constraints limited differentiation between similarly ranked bridges, particularly for customer and freight impacts.

Overall assessment

- **The prioritisation framework remains conceptually sound and directionally robust.**
- **Data availability and integration — not framework design — are the primary constraints on confidence and scalability.**
- **Targeted data uplift and integration with TfNSW asset information systems (such as TACP) is required to embed the framework into business-as-usual planning and governance processes.**

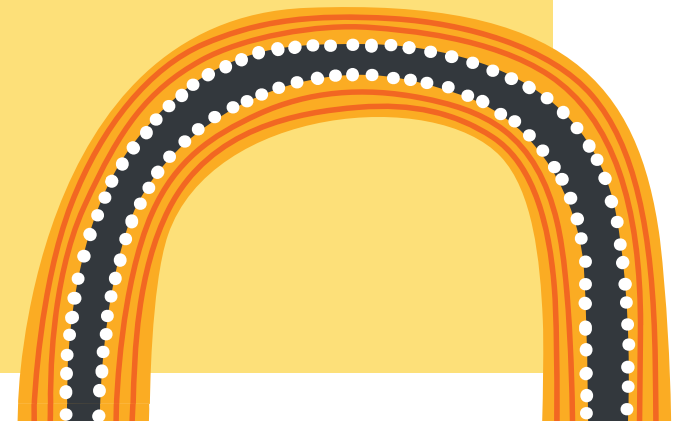
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INTERIM PRIORITISATION TOOL AND RESULTS



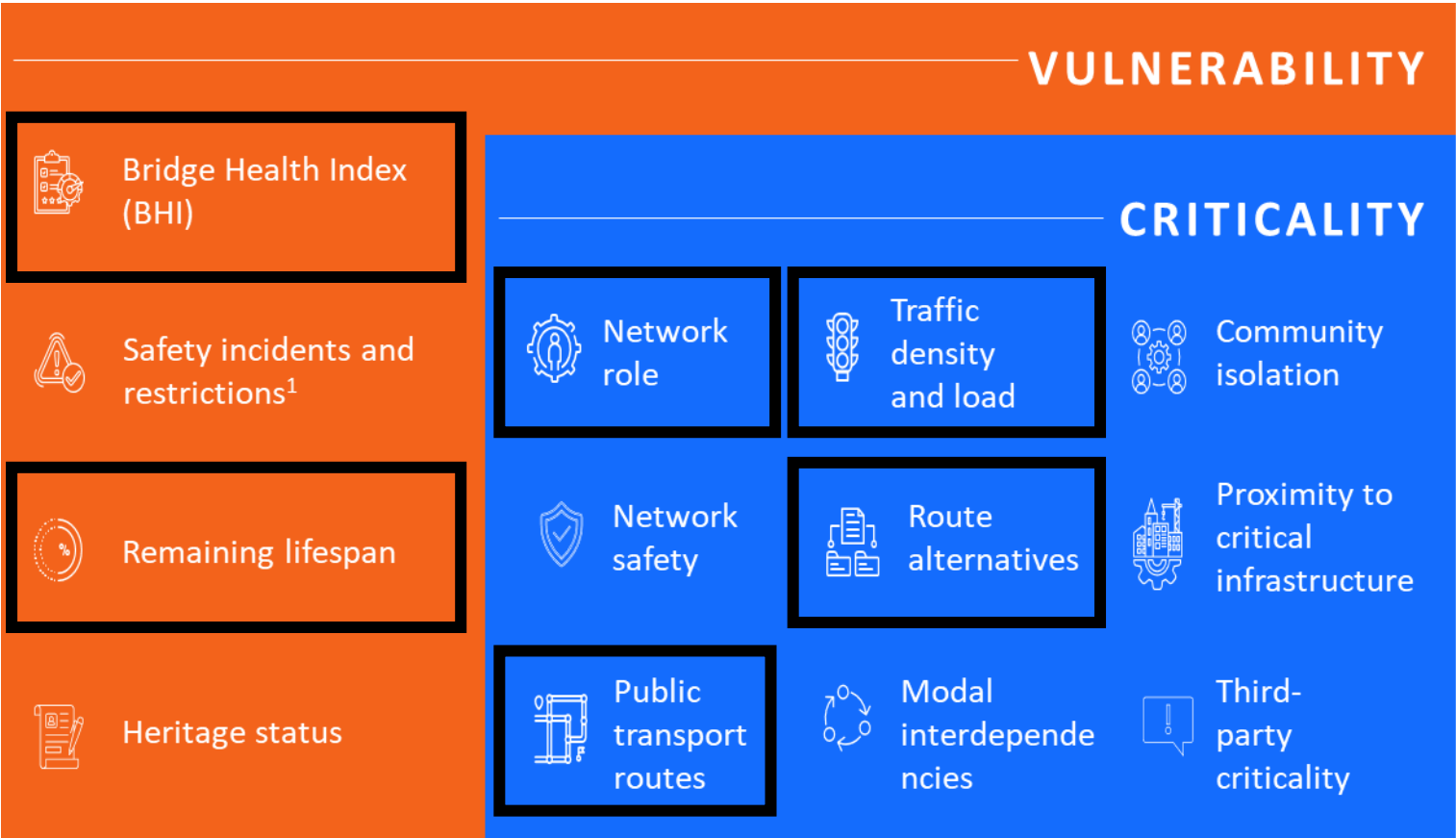
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INTERIM PRIORITISATION CRITERIA



Bridge prioritisation criteria for interim prioritisation

Informed by the availability and quality of data observed (as noted in Section 4), the interim prioritisation tool (to automate and standardise the prioritisation process) was developed with the available data, its veracity and completeness across the population of bridges to be prioritised. The below criteria were included in the interim prioritisation tool design.



CRITICALITY – *asset function*
Customer and service outcomes, enablement

VULNERABILITY – *asset condition*
Asset status, health and condition measures

Criteria to be applied on current bridge asset register due to data limitations

1. This criterion was renamed and its definition amended following feedback on the Pilot Phase outputs. The previous criterion “Historical safety incidents” is included in the Appendix material.

Interim prioritisation criteria weighting

Condition emphasis and weighting adjustment

Weightings for the scored criteria and contributing user types were adjusted to reflect the below principles:

- Bridges in ‘Poor’ condition to be prioritised
The **Vulnerability** criterion ‘*Bridge Health Index (BHI)*’ was weighted such that its contribution is equivalent to **five-times** any other individual criterion included in the interim prioritisation.
i.e., $\text{Weighting}_{\text{BHI}} = 7.50$
- Each **Criticality** criterion contribute equally
Each **Criticality** criterion was weighted such that its contribution is equivalent to any other individual **Criticality** criterion.
Active Transport user-type contributions were down-scaled, reflecting data confidence and tendency to dominate prioritisation scores

Cell values = $\text{Weighting}_{\text{UserType}} \times \text{Weighting}_{\text{Criterion}}$

	USER TYPES	Road - Passenger	Road – Heavy vehicles Class 2 & 3	Active Transport - Pedestrians	Active Transport - Cyclists	Rail – Passenger	TOTAL CRITERION WEIGHTING
CRITERIA	Weighting Applied	1.00	1.00	0.50	0.50	1.00	
Network role	1.50	1.50	N/A	N/A	N/A	N/A	1.50
Public transport routes	0.75	0.75	N/A	N/A	N/A	0.75	1.50
Traffic density and load	1.50	1.50	N/A	N/A	N/A	N/A	1.50
Route alternatives	0.50	0.50	0.50	0.25	0.25	N/A	1.50
Bridge Health Index (BHI)	7.50	N/A – Scored by bridge, not user type					7.50

Interim prioritisation criteria mapping

Criticality – Route alternatives

Data processing and mapping approaches taken for the interim prioritisation are outlined by user type in the table below

User Type	Data Source	Method
Road – Passenger Active Transport – Pedestrians Active Transport – Cycling	Open Street Maps network	1. Data cleaning / filter on bridges to remove wildlife crossings etc. 2. Match bridge lat-long (GPS coordinates) to closest network edge 3. Determine if the bridge can be used by the user type 4. Shortest-path algorithm along edges available to that user-type (excluding the bridge itself) 5. Estimate time of new route 6. Manual review of a sample of results
Road – Heavy vehicles Class 2 & 3	- Open Street Maps network - NHVR map of where representative vehicle types permitted	1. Choose representative vehicle(s)¹ 2. Select NHVR network map of where that vehicle is permitted 3. Apply shortest path algorithm, as above

1. TfNSW freight stakeholders provided input for the selection of the representative vehicles.
The selected vehicles were “*PBS 2b*”-type and “*PBS 3a*”-type vehicles.

Interim prioritisation criteria mapping

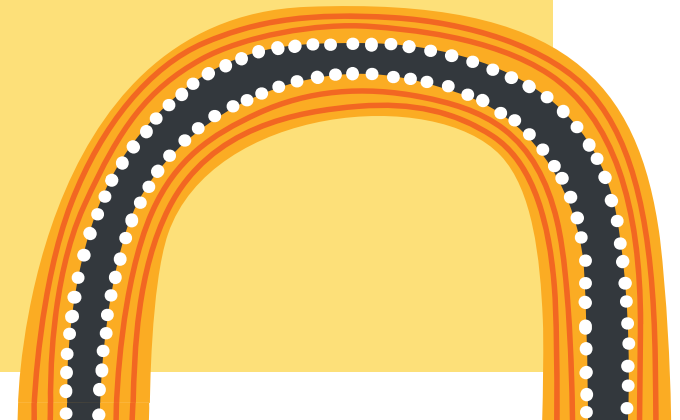
Criticality – Public transport routes

Data processing and mapping approaches taken for the interim prioritisation are outlined by user type in the table below

User Type	Data Source	Method
Road – Passenger	- Transport Open Data Hub: NSW Bus Routes - NHVR map (routes where buses permitted)	1. Count the services passing over / under the bridge
Rail – Passenger	- TrainLink timetable - Transport Open Data Hub: Sydney Train Routes	1. Count the routes passing over / under the bridge

04.b

INTERIM HURDLE RATES



Multi criteria analysis

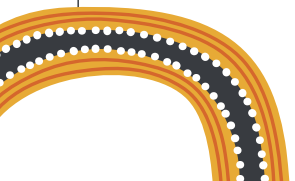
Hurdle rates and data sources – Interim Prioritisation

The hurdle rates and data sources on the following slides reflect those applied in the **Interim Prioritisation (based on data limitations)**

Hurdle rates have only been listed on the subsequent slides for MCA criteria and user types applied in the interim prioritisation output; non-applied user type are greyed out with *light grey* fill, with non-applicable user types for certain criteria denoted with “N/A” and in *dark grey* fill.

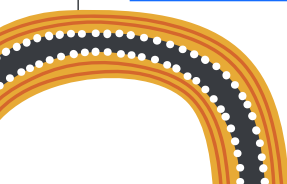
Those criteria not applied in the interim prioritisation output are not included.

Further refinement of the hurdle thresholds and inclusion of additional user types (and criteria) should occur after Transport’s data sources are further developed and the quality of the data is aligning with the requirements of the prioritisation framework.



Criticality Criteria and Hurdle Rates

Criteria	User types	5	4	3	2	1	0
Network role <i>Importance of the bridge in the broader transport network. Reflects the role of the bridge in maintaining national, state and local transport continuity. Includes if the bridge forms part of the National Land Transport Network, a critical freight or passenger corridor, or a Strategic Cycleway Corridor.</i>	Road – passenger vehicles & light commercial vehicles	Part of <i>National Land Transport Network</i>	On a <i>State Road</i>	On a <i>Regional Road</i>	N/A	On a <i>Local Road</i>	Local access only
	Road – Class 2 & 3 HV						
	Road – Class 1 HV						
	Rail – passenger trains						
	Rail – freight trains						
	Active transport – pedestrians						
	Active transport – cyclists						
	Water transport – commuter ferries and private vessels						
	Water transport – freight shipping and other commercial vessels						



Criticality

Vulnerability

Network role

Network safety

Public transport routes

Traffic density

Route alternatives

Modal interdependencies

Proximity to critical services

Third-party criticalities

Community isolation

Bridge Health Index (BHI)

Safety incidents and restrictions

Remaining lifespan

Heritage status

Criticality Criteria and Hurdle Rates

Criteria	User types	5	4	3	2	1	0
Public transport routes <i>The bridge's contribution to public transport continuity, including available seat kilometres, school bus routes, and intermodal connectivity</i>	Road – passenger vehicles & light commercial vehicles	Supports 50+ bus routes (public and school bus) per day (Mon-Fri)	Supports 30 - 49 bus routes (public and school bus) per day (Mon-Fri)	Supports 15 - 29 bus routes (public and school bus) per day (Mon-Fri)	Supports 5 - 14 bus routes (public and school bus) per day (Mon-Fri)	Supports 1 - 4 bus routes (public and school bus) per day (Mon-Fri)	Bridge does not support any bus routes (public and school bus) per day (Mon-Fri)
	Road – Class 2 & 3 HV	N/A	N/A	N/A	N/A	N/A	N/A
	Road – Class 1 HV	N/A	N/A	N/A	N/A	N/A	N/A
	Rail – passenger trains	Supports 5+ passenger train routes	Supports 4 passenger train routes	Supports 3 passenger train routes	Supports 2 passenger train routes	Supports 1 passenger train route	Does not support any passenger train routes
	Rail – freight trains	N/A	N/A	N/A	N/A	N/A	N/A
	Active transport – pedestrians	N/A	N/A	N/A	N/A	N/A	N/A
	Active transport – cyclists	N/A	N/A	N/A	N/A	N/A	N/A
	Water transport – commuter ferries and private vessels						
	Water transport – freight shipping and other commercial vessels	N/A	N/A	N/A	N/A	N/A	N/A

Criticality

Vulnerability

Network role

Network safety

Public transport routes

Traffic density

Route alternatives

Modal interdependencies

Proximity to critical services

Third-party criticalities

Community isolation

Bridge Health Index (BHI)

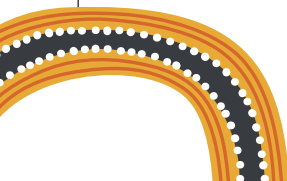
Safety incidents and restrictions

Remaining lifespan

Heritage status

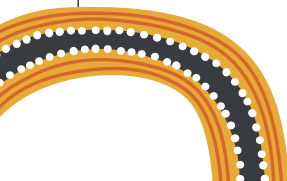
Criticality Criteria and Hurdle Rates

Criteria	User types	5	4	3	2	1	0
Traffic density and load <i>Volume and loading (damage) of traffic using the bridge.</i> Note: <i>This should be the traffic volume / loading traversing the bridge (over or under) in normal operating conditions.</i> <i>For bridges with access restrictions in effect:</i> <i>If the access restriction has been in effect for <6mo: apply the un-restricted traffic volume / loading</i> <i>If the access restriction has been in effect for 6-18mo: apply the average of the un-restricted and the restricted traffic volume / loading</i> <i>If the access restriction has been in effect for >18mo: apply the restricted traffic volume / loading</i>	Road – passenger vehicles & light commercial vehicles	35,000+ AADT	22,500 – 34,999 AADT	12,500 – 22,499 AADT	6,500 – 12,499 AADT	2,500 – 6,499 AADT	Less than 2,500 AADT
	Road – Class 2 & 3 HV						
	Road – Class 1 HV						
	Rail – passenger trains						
	Rail – freight trains						
	Active transport – pedestrians						
	Active transport – cyclists						
	Water transport – commuter ferries and private vessels						
	Water transport – freight shipping and other commercial vessels						



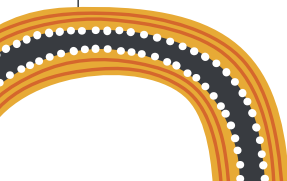
Criticality Criteria and Hurdle Rates

Criteria	User types	5	4	3	2	1	0
Route alternatives <i>Availability of viable alternative routes or crossings if the bridge becomes unavailable, and the delay imposed (detour time).</i>	Road – passenger vehicles & light commercial vehicles	No alternative route, or Detour time 180+ min	Detour time 90 – 179 min	Detour time 45 – 89 min	Detour time 22.5 – 44 min	Detour time 7.5 – 22 min	Detour time less than 7.5 min
	Road – Class 2 & 3 HV	No alternative route, or Detour time 180+ min	Detour time 90 – 179 min	Detour time 45 – 89 min	Detour time 22.5 – 44 min	Detour time 7.5 – 22 min	Detour time less than 7.5 min
	Road – Class 1 HV						
	Rail – passenger trains						
	Rail – freight trains						
	Active transport – pedestrians	No alternative route, or Detour time 90+ min	Detour time 45 – 89 min	Detour time 25 – 44 min	Detour time 15 – 24 min	Detour time 5 – 14 min	Detour time less than 5 min
	Active transport – cyclists	No alternative route, or Detour time 90+ min	Detour time 45 – 89 min	Detour time 25 – 44 min	Detour time 15 – 24 min	Detour time 5 – 14 min	Detour time less than 5 min
	Water transport – commuter ferries and private vessels						
	Water transport – freight shipping and other commercial vessels						



Vulnerability Criteria and Hurdle Rates

Criteria	5	4	3	2	1	0
Bridge Health Index (BHI) <i>A broad qualitative indicator of potential risk of structural failure derived from the condition ratings of the bridge elements from the latest inspection of a bridge</i>	Poor	N/A	Fair	N/A	Good	As Built



Criticality

Network role

Network safety

Public transport routes

Traffic density

Route alternatives

Modal interdependencies

Proximity to critical services

Third-party criticalities

Community isolation

Vulnerability

Bridge Health Index (BHI)

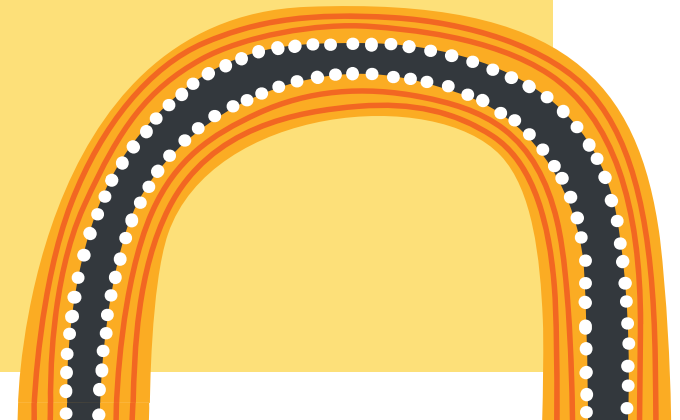
Safety incidents and restrictions

Remaining lifespan

Heritage status

04.c

RESULTS



TfNSW Bridge AM Assurance and Prioritisation



Bridge Prioritisation Dashboard with real data points (from TACP)

Region

- ☐ NORTH REGION
- ☐ SOUTH REGION
- ☐ SYDNEY REGION
- ☐ WEST REGION

Owner

- ☐ A.C.T. Government
- ☐ A.R.T.C
- ☐ Contractor to TfNSW
- ☐ Council (LGA)
- ☐ Country Rail Contracts

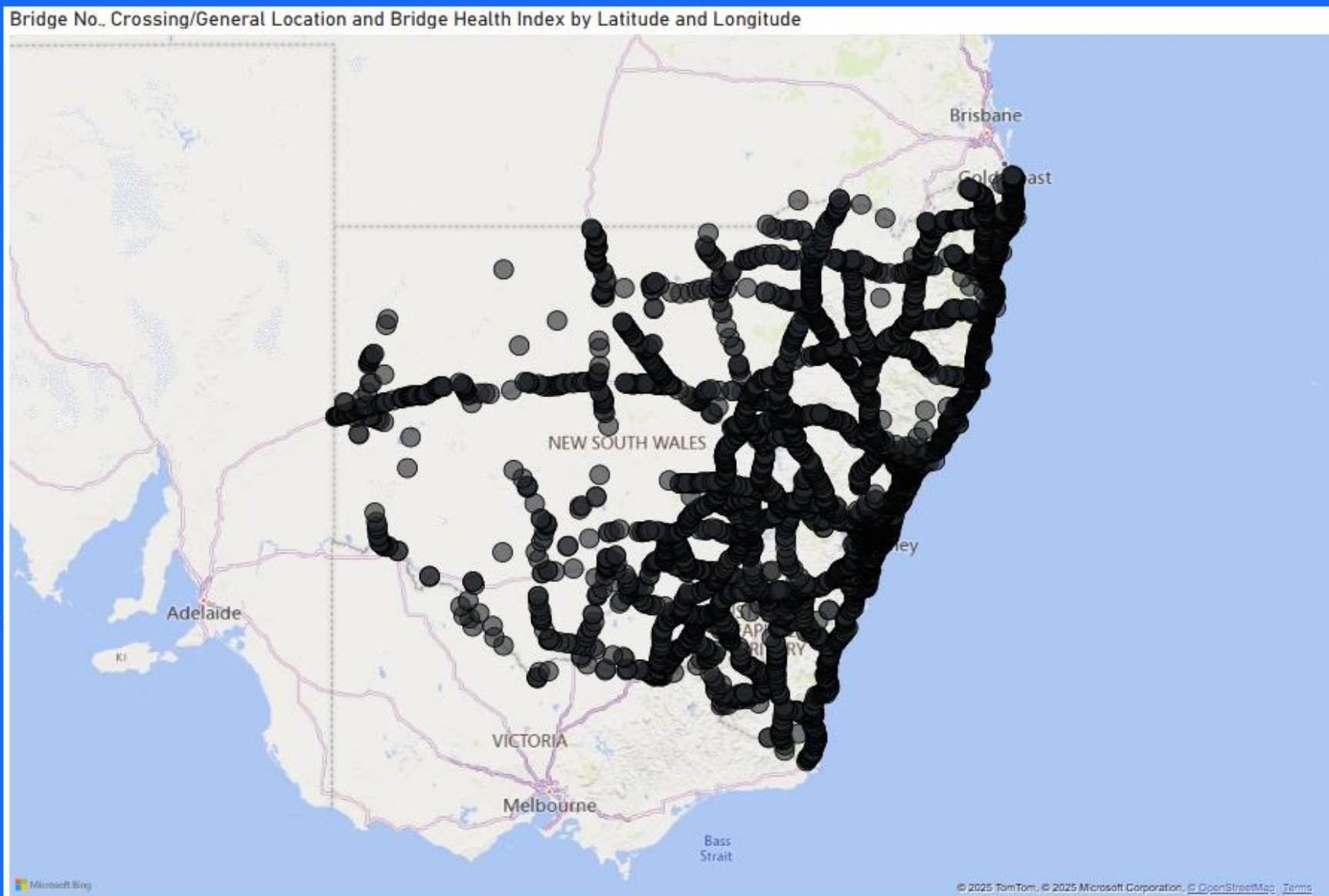
Maintainer

- ☐ A.C.T. Government
- ☐ A.R.T.C
- ☐ Contractor to TfNSW
- ☐ Council (LGA)
- ☐ Country Rail Contracts

Bridge Number

Search

Bridge No.	Description
1	ON-LOAD RAMP
2	EB BRIDGE ON HIGH ST
3	WB BRIDGE ON HIGH ST
4	BRIDGE ON MOUNT ST
5	BRDG ON MOUNT ST
6	PED BR AT RIDGE ST
9	BRIDGE ON ERNEST ST
10	BRIDGE ON ERNEST ST
11	BRIDGE ON ERNEST ST



Operational Status

- ☐ Closed To Traffic
- ☐ Completed but Not Opened To Tra
- ☐ Decommissioned (Not controlled ...
- ☐ Opened To Traffic

Material or drag fields to populate this v

- Timber
- Concrete
- Steel

Inspection Status or drag fields to populate this v

- Completed
- Scheduled
- Overdue

Latest Inspection Date populate

Start Date vis End Date

Bridge Health Index

- ☐ AS-BUILT
- ☐ FAIR
- ☐ GOOD
- ☐ POOR

Load Type or drag fields to populate this v

- Road
- Rail
- Pedestrian

Specialisation Category or drag fields to populate this v

- Heritage
- Toll
- Other

43

Interim bridge prioritisation output – Top 20

Tables below reflect the results of the top 20 prioritised bridges from the interim tool (using the criteria and weightings in Section 4a). Comparison is made to the original TfNSW Ranked Bridge list from March 2025. Resulting Vulnerability and Criticality scores are shown against each bridge.

RANK		Vulnerability Score	Criticality Score	Ranking in Original TfNSW List
1	B1518 Forster/Tuncurry Bridge	37.50	16.75	11
2	BRIDGE ON SH1 OVER F.6 AT GHOSTS CREEK GHOSTS CREEK BRIDGE	37.50	16.25	NEW
3	SBEAM BRIDGE WYONG RIVER 19.66K N OF GOSFORD WYONG RIVER	37.50	14.50	NEW
4	BRIDGE MACDONALDS CREEK 12.22KM N MUDGEES	37.50	14.25	NEW
4	BRIDGE OVER RAILWAY AT THIRROUL	37.50	14.25	NEW
6	B1365 Swansea north bound	37.50	14.00	10
7	BRIDGE OVER TUROSS RIVER 1.6KM NORTH OF BODALA	37.50	14.00	NEW
8	6 CELL CULVERT SCOPES RANGE CREEK 104.0KM W WILCANNIA	37.50	13.75	NEW
9	STRUS BRIDGE ON SH10 NAMBUCCA RIVER AT MACKSVILLE MACKSVILLE BRIDGE	37.50	13.75	NEW
10	BRIDGE ON SR63 OVER GWYDIR RIVER 0.87KM NTH BINGARA CAMPBELL BRIDGE	37.50	13.50	NEW

RANK		Vulnerability Score	Criticality Score	Ranking in Original TfNSW List
11	CBEAM BRIDGE (AB1) KRUI RIVER 25.08K W OF MERRIWA COLLARROY BRIDGE	37.50	13.25	NEW
11	Rocket Street Overbridge Bathurst	37.50	13.25	NEW
11	BRIDGE RAILWAY BATHURST ROCKET STREET BRIDGE ¹	37.50	13.25	NEW
14	3 CELL CULVERT UNNAMED CREEK 5.4KM W EUGOWRA	37.50	13.00	NEW
15	CULVERT ON SH12 CHINAMANS GULLY NR MATHESON 29.78KM W GLN INN	37.50	12.75	NEW
15	5/1.5X0.9 CULV ON SH11 UNNAMED CREEK 28.24KM WEST GUNN	37.50	12.75	NEW
15	BRIDGE OVER SAUCY CREEK	37.50	12.75	NEW
15	S/B F6 BRIDGE OVER MR295 AT UNANDERRA	37.50	12.75	NEW
19	B6168 Wallaga Lake Bridge	37.50	12.50	22
19	BRIDGE OVER S/B DUCK CREEK ON SH1 AT MT. BROWN	37.50	12.50	NEW
19	SGIRD BRIDGE DORA CK 6.29K E OF MORRISET	37.50	12.50	NEW
22	B2322 Old Grafton Bridge	37.50	12.25	14
22	12 CELL CULVERT MYALL VALE CHNL 29.95KM W NARRABRI	37.50	12.25	NEW

1. This bridge entry refers to the same structure as the entry above ('Rocket Street Overbridge Bathurst'); however, they have unique TACP Asset ID numbers

Evaluation of results against original TfNSW prioritisation output

Factors influencing a change in ranking for the Top 15 bridges

Ranking in Original TfNSW List		Vulnerability Score	Criticality Score	New Rank	Factors influencing new ranking
1	B98 Bentley's Bridge	37.50	6.25	66	- No traffic loading data available - No public transport routes identified automatically; manual review suggests at least 2 public bus routes present - Short detour times for present user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
2	Bridge 333 over Nepean River	37.50	7.75	59	(This is the same bridge as "B333 Victora Bridge" below)
3	B333 Victoria Bridge	37.50	7.75	59	- No traffic loading data available - No public transport routes identified automatically; manual review suggests at least 4 public bus routes present - Short detour times for some user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
4	B390 Galston Gorge Bridge	37.50	1.25	96	- No traffic loading data available - No public transport routes identified automatically - Road – Passenger users not automatically identified; no detour time calculated - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
4	B413 St Albans Bridge	37.50	9.75	44	- No traffic loading data available - No public transport routes identified automatically - Significant detour times for identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
6	B875 Hampden Bridge, Kangaroo Valley	37.50	8.75	51	- Traffic loading data available, low volume - Public transport routes identified automatically - Significant detour times for some identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
7	Bridge 941 and 942 road over rail bridge in Wollongong leading into the port	37.50	10.50	34	- Traffic loading data available, mid-low volume - Public transport routes identified automatically - Significant detour times for some identified user types - On NHVR map for one representative Class 2 vehicle
8	Bridge 941 and 942 road over rail bridge in Wollongong leading into the port	37.50	10.50	34	(This is the same bridge as "B942 road over rail bridge in Wollongong" above)
9	B942 JJ Matthews Bridge	37.50	10.50	34	(This is the same bridge as "B942 road over rail bridge in Wollongong" above)

Evaluation of results against original TfNSW prioritisation output

Factors influencing a change in ranking for the Top 15 bridges

Ranking in Original TfNSW List		Vulnerability Score	Criticality Score	New Rank	Factors influencing new ranking
10	B1365 Swansea north bound	37.50	14.00	6	- Traffic loading data available, mid-high volume - Public transport routes identified automatically - Significant detour times for some identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
11	B1518 Forster/Tuncurry Bridge	37.50	16.75	1	- Traffic loading data available, mid-high volume - Public transport routes identified automatically, high count of bus lines - Significant detour times for some identified user types - On NHVR map for one representative Class 2 vehicle
12	B1557 Kayuga Bridge	37.50	6.75	63	- Traffic loading data available, mid-low volume - Public transport routes identified automatically - Significant detour times for some identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
13	B1752 Clarencetown Bridge	37.50	8.25	56	- Traffic loading data available, very low volume - Public transport routes identified automatically - Significant detour times for some identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated
14	B2322 Old Grafton Bridge	37.50	12.25	22	- Traffic loading data available, mid-high volume - Public transport routes identified automatically, very high count of bus lines - Significant detour times for some identified user types - Not on NHVR map for representative Class 2 vehicles; no detour time calculated - Local road, not State or National Land Transport Network road
15	B3215 Swan Hill Bridge	37.50	11.00	32	- Traffic loading data available, very low volume - Public transport routes identified automatically 'Infinite' detour times for all identified user types - On NHVR map for representative Class 2 vehicles; 'infinite' detour time calculated

05

HIGH LEVEL IMPROVEMENT ROADMAP



Improvement Recommendations and Roadmap

The following improvement recommendations have been identified as high priority from the development of the bridge prioritisation framework and interim tool. The recommendation have been sequenced into a roadmap for implementation, focusing on first on the foundational data improvements needed then extension of data requirements to inform improved bridge prioritisation.

0 – 6 Months: Critical Foundational Data Improvements

ID	Item	Description
A1	TACP asset data quality improvement	Ensure that all bridges in TACP have the following data completed in accordance with the Attribute Definition in the TfNSW ACS: • Constructed Date and Commissioned Date • Associated GPS coordinates assigned • Asset condition (note dependency on Item A2 below) • Heritage Inventory Number
A2	Develop consistent bridge condition rating in TACP	• Establish a consistent condition rating methodology for all bridges (across all modes) • Evaluate all bridges with the standardised rating methodology • Update the bridge condition ratings and date of rating in TACP
A3	Capture current bridge restrictions across transport network	• Establish centralised reporting framework for all current bridge restriction covering load and dimensional limits • Reporting to include details as when introduced, the cause (and required intervention) of the restriction and assessment of impact on bridge users • This data will inform the Safety Incidents and Restrictions criteria in the bridge prioritisation framework

Improvement Recommendations and Roadmap

The following improvement recommendations have been identified as high priority from the development of the bridge prioritisation framework and interim tool. The recommendation have been sequenced into a roadmap for implementation, focusing on first on the foundational data improvements needed then extension of data requirements to inform improved bridge prioritisation.

0 – 6 Months: Critical Foundational Data Improvements

ID	Item	Description
A4	Capture of rail traffic density and load	Extend the current interim bridge prioritisation tool to incorporate • Use of ARTC and CRN tonnage and train counts data • SWTT for passenger rail services This data will need to be validated with the improved geospatial data in TACP (refer to item A1) to map location of bridges to rail corridors (in accordance with the train count data).

Improvement Recommendations and Roadmap

The following improvement recommendations have been identified as high priority from the development of the bridge prioritisation framework and interim tool. The recommendation have been sequenced into a roadmap for implementation, focusing on first on the foundational data improvements needed then extension of data requirements to inform improved bridge prioritisation.

6 – 12 Months: Extend Asset Data Requirements

ID	Item	Description
B1	Geolocate each asset (GPS coordinates)	• Extend current geospatial attributes in TACP to assign a GPS coordinate for the start and end point of each bridge • This will enable more accurate location analysis to eliminate challenges with bridge identification where there are multiple bridges in one location
B2	Develop a TfNSW Asset Criticality Framework (economic)	• Provide clear understanding of transport users by corridor to help inform bridge users for prioritisation framework • Asset criticality rating methodology to be based on economic contribution of assets to transport network and corridors • Establish asset criticality rating that can be used consistently across all modes of transport and corridor asset classes and update “Asset Criticality Code” in TACP • This will inform the bridge user types for each bridge, Network Role, Modal Interdependencies and Network Safety criterion in the bridge prioritisation framework
B3	Capture of road freight use for all heavy vehicles	Improve the quality of freight road user traffic density and load through consideration of TfNSW’s Freight Data Hub which publishes freight and traffic-related datasets including heavy vehicle statistics, traffic count information and telematics-based data relevant to high-productivity vehicles. This data will need to be validated using methods such as results from TfNSW’s machine learning dashboard with cameras and aligned with access management arrangements through consolidating multiple proxy data sources (e.g. NHVR permits and policy data). This will have dependency with Item B2.

Improvement Recommendations and Roadmap

The following improvement recommendations have been identified as high priority from the development of the bridge prioritisation framework and interim tool. The recommendation have been sequenced into a roadmap for implementation, focusing on first on the foundational data improvements needed then extension of data requirements to inform improved bridge prioritisation.

6 – 12 Months: Extend Asset Data Requirements

ID	Item	Description
B4	Understanding of third-party critical service on bridges	Develop and define a data attribute for bridges that can identify the quantum of third-party services (e.g., utilities, pipelines, communication assets) that the bridge supports. This data will need to be sourced from detailed asset bridge audits and/or local information managed by bridge maintainers.
B5	Capture of geospatial proximity of each bridge to critical infrastructure and community access	Develop and define the following data attributes for each bridge to be captured through geospatial analysis: • Proximity of each bridge to critical infrastructure (bridge's role in maintaining access to essential and emergency services) • Proximity of each bridge to regional centres that can be isolated if a bridge fails or access is restricted.
B6	Expand ACS Standard to include specialised bridge attributes for prioritisation	Expand the Data Dictionary in the ACS to define and standardise bridge specialised attributes to capture the relevant data requirements as noted in Items B1, B4 and B5

Improvement Recommendations and Roadmap

The following improvement recommendations have been identified as high priority from the development of the bridge prioritisation framework and interim tool. The recommendation have been sequenced into a roadmap for implementation, focusing on first on the foundational data improvements needed then extension of data requirements to inform improved bridge prioritisation.

12 – 24 Months: Data Embedment and Integration

ID	Item	Description
C1	Develop and implement centralised corridor demand and traffic management system	TfNSW to invest in an integrated system that centralises and governs corridor demand, traffic and asset-use data across TfNSW and external source systems, to provide the data and modelling needed to understand where demand, congestion, incidents, freight movements and asset performance occur on the network. For example, this can include an integrated systems solution covering: • System of record for integrated corridor demand / traffic / telemetry / asset-use analytics (with interface with TACP) providing: • corridor demand history and forecasting, real-time traffic and freight data ingestion, cross-modal and cross-agency data integration, self-service dashboards for planners, asset managers and corridor operators, asset-use metrics such as corridor loads, utilisation, reliability, delay exposure and demand-growth hotspot. • Spatial network / corridor / route-event management layer providing: • which corridor segments carry the highest freight volumes • where congestion and incident patterns are concentrated • which bridges and corridor assets are exposed to the greatest heavy-vehicle loading • how route restrictions, works and detours shift demand across the network

06

APPENDIX

6.1 Interim Prioritisation Findings and Results (December 2025)

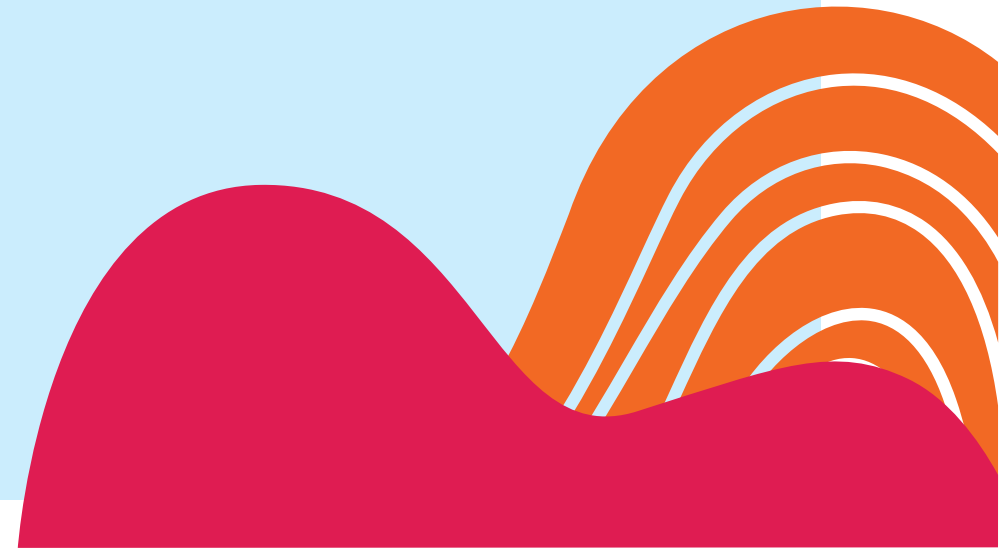
6.2 Updated Interim Prioritisation Findings and Results (February 2026)

6.3 Proposed Bridge Prioritisation Dashboard



6.1

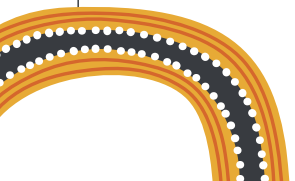
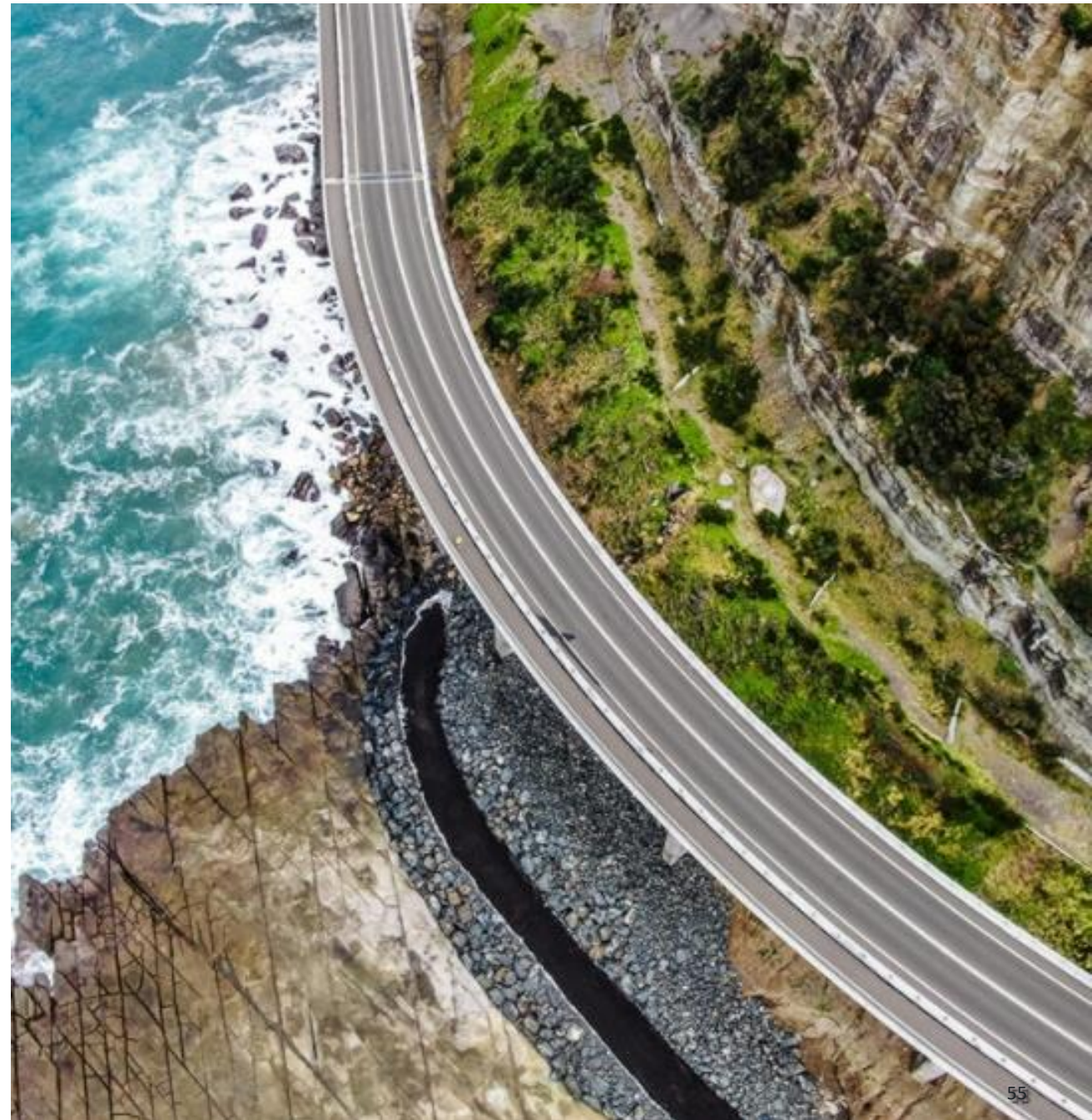
Interim Prioritisation Findings and Results (December 2025)



Interim Prioritisation

The interim prioritisation was undertaken on the 99 Vulnerable Bridges Sample Set (from preliminary analysis). Key items of note:

- All prioritisation criteria are equally weighted which can be adjusted following review of results in finalising the prioritisation model (next phase)
- Analysis demonstrated the draft prioritisation model produces results aligned with priority ranking of bridges based on transport outcomes
- Analysis validated required data sources from current data limitations to inform finalisation of prioritisation model (next phase) for application to bridge portfolio
- Interim prioritisation results represent assessment of 1600 data points across 99 bridges for the 13 prioritisation criteria across 14 user types



Findings and limitations: Vulnerability Criteria

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Bridge Health Index (BHI)	Medium	Bridge health criteria and data not common across different modes (road and rail use different measures of asset condition)	Project Team have engaged TfNSW Chief Engineer to determine a common bridge condition measure applicable to all modes
Historical safety incidents	Low	Historical safety incident data was not available for this analysis and was not applied to the interim prioritised list	Project team to identify relevant sources of bridge safety incident data with asset stewards that can be attributed to individual bridges
Remaining lifespan	Medium	Remaining useful life data was not available for the rail bridges in TACP (available for road bridges)	Project team to identify relevant sources of remaining useful data with Country Rail asset steward, including data update for TACP
Heritage status	High	Bridges were identified on the relevant Australian and State Heritage Registers	

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Findings and limitations: Criticality Criteria

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Network role	Medium	Limitations in classifying heavy vehicle classes (only class 1 identified from NHVR). Active transport use and under bridge use determined using Google Maps.	Project team to work with Freight team to source more accurate heavy vehicle classification and identify more efficient / reliable GIS/database options available with TfNSW stakeholders for under bridge use.
Network safety	Low	Network safety data was not available for this analysis and was not applied to the interim prioritised list	Project to identify with TfNSW stakeholders relevant data source for bridge network safety use, including emergency management and evacuation functions.
Public transport routes	Medium	Google Maps and public transport network maps were sourced to identify number of public transport services over each bridge	Project to identify with TfNSW stakeholders more relevant and accurate data source for bridge's contribution to public transport service such as available seat kilometres, school bus routes, and intermodal connectivity.
Traffic density and load	Medium	AADT and BIS data was used to determine traffic volumes and heavy vehicle loads for road bridges. SWTT was used for passenger train volumes, and data was not available for freight train volumes.	Project to identify with TfNSW stakeholders more relevant and accurate data source for heavy vehicle loads (ESAL), passenger/freight rail volume and activity transport traffic for each bridge.
Route alternatives	Medium	Google Maps was sourced to identify and calculate route alternatives to determine total detour time for each bridge for passenger and active transport.	Project to identify with TfNSW stakeholders more relevant and accurate traffic modelling / data source for calculating route alternatives for all transport users.

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Findings and limitations: Criticality Criteria (cont.)

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

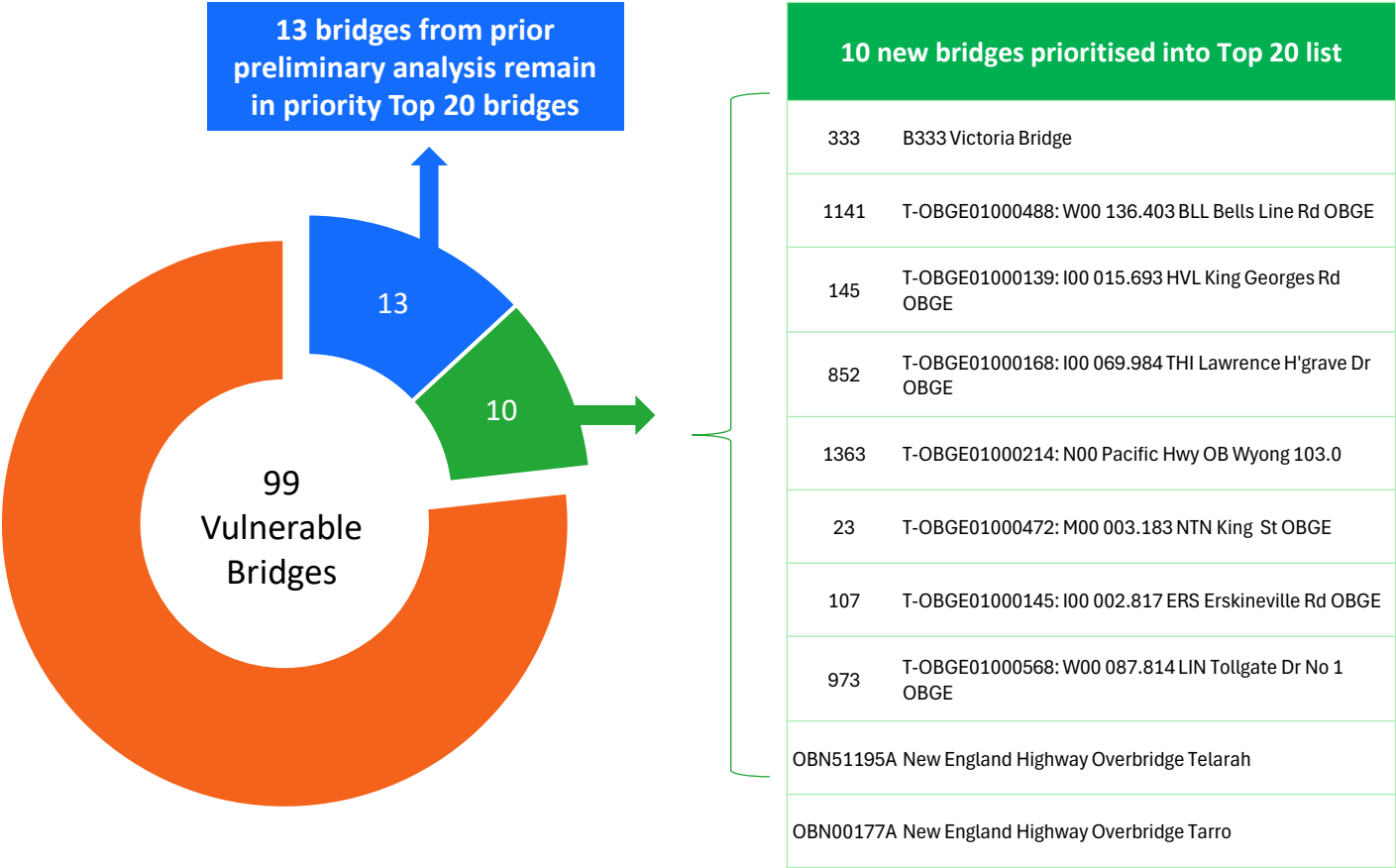
Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Modal inter-dependencies	Low	Modal interdependency data was not available for this analysis and was not applied to the interim prioritised list	Project to identify with TfNSW stakeholders relevant data source to assess how the failure of a bridge may contribute to the mode shift of users and relationship/applicability with the route alternative metric.
Proximity to critical services	Medium	Identification of critical services / infrastructure supported by each bridge was conducted using NSW Open Map Data	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for identification of critical services
Third-party criticality	Medium	Identification of third-party assets on bridges was conducted using Google Maps for each individual bridge	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for identification of third-party assets
Community isolation	Medium	Identification population that is isolated if a bridge fails was conducted using Google Maps and ABS population data for each individual bridge	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for assess community/population isolation impacts (where no alternative routes are available)

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Top 20 Priority Bridges

- From the interim prioritised list of the Top 20 ranked bridges (23 in total as 3 bridges have equal scoring), 13 remain the same as in the initial assessment, while 10 new bridges are part of the Top 20.
- Key drivers influencing the additional 10 bridges as priorities:
 - Consideration of a waterway needs and multiple customer user groups
 - Limited alternative routes, with long detours required
 - Vulnerability due to heritage status, remaining lifespan, and third-party critical assets on bridges



Interim Prioritisation Results

Greater Sydney		Vulnerability Subtotal	Criticality Subtotal	Grand Total	"New" Rank	Comments
B98	Bentley's Bridge, Elizabeth Bay	11	25	36	25	Outside Top 21 ranking: mid-level heritage rating, no user types under bridge
B44	Pacific Hwy, Turrumunga OBGE	4	40	44	12	Still in Top 21 ranking
B12555	Meadowbank footbridge	14	37	51	2	Still in Top 21 ranking
Regional South						
B938	Springhill Road, Port Kembla	3	19	22	73	Outside Top 21 ranking: missing Vulnerability and freight rail volume data
B941	JJ Mathews bridges, Wollongong	5	33	38	21	Still in Top 21 ranking
B942	JJ Mathews bridges, Wollongong	7	33	40	16	Still in Top 21 ranking
B3244	Tooleybuc Bridge	11	27	38	21	Still in Top 21 ranking
B6298	Sheehan Bridge, Hume Hwy	4	41	45	9	Still in Top 21 ranking
B3215	Swan Hill	11	29	40	16	Still in Top 21 ranking
B875	Hampden Bridge	11	21	32	34	Outside Top 21 ranking: no heavy vehicle access, no user types under bridge
Regional West						
B972	Great Western Hwy, Linden	3	42	45	9	Still in Top 21 ranking
B4122	Angus Bridge, Mitchell Hwy	5	22	27	55	Outside Top 21 ranking: limited Vulnerability and freight rail volume data, no user types under bridge
CRN	Lachlan Valley Way, Yass Junction	1	27	28	49	Outside Top 21 ranking: missing Vulnerability and freight rail/road volume data
B4767	Boothaguy Creek Bridge	3	22	25	60	Outside Top 21 ranking: limited Vulnerability and freight rail volume data, no user types under bridge
B4660	Scabbing Flat Bridge	10	23	33	69	Outside Top 21 ranking: no user types under bridge, no heritage rating
Regional North						
B1365	Swansea northbound	8	43	51	2	Still in Top 21 ranking
CRN	Industrial Drive, Port Waratah	1	32	33	31	Outside Top 21 ranking: limited Vulnerability and freight rail volume data
CRN	Maud Street, Sandgate	3	41	44	12	Still in Top 21 ranking
CRN	Church Street, Maitland	5	38	43	15	Still in Top 21 ranking
B1518	Forster / Tuncurry	7	47	54	1	Still in Top 21 ranking
B2322	Old Grafton bridge	13	26	39	18	Still in Top 21 ranking

Interim Prioritisation Results

Top 10 priority bridges from 99 Vulnerable Bridges Sample Set based on results of multi-criteria analysis

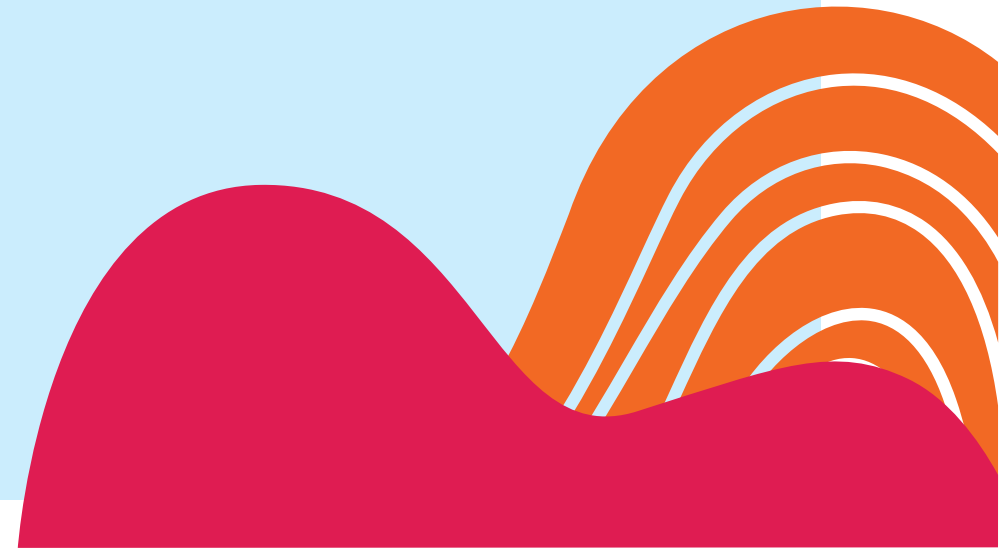
RANK	Bridge Number	Bridge Name	Vulnerability Score	Number of User Types Score (On Under)	Network Role Score	Public Transport Routes Score	Route Alternatives Score	Proximity to Critical Infrastructure Score	Bridge Over Water Transport
1	1518	B1518 Forster/Tuncurry Bridge	7	6 (4 2)	11	4	19	5	Yes
2*	1365	B1365 Swansea north bound	8	6 (4 2)	11	3	14	5	Yes
2*	12555	B-12555 - Footbridge connecting Meadowbank and Rhodes	14	5 (3 2)	11	8	12	0	Yes
4*	OBN00177A	New England Highway Overbridge Tarro	1	6 (4 2)	19	5	8	4	No
4*	333	B333 Victoria Bridge	13	4 (2 2)	6	4	17	3	Yes
6	1141	T-OBGE01000488: W00 136.403 BLL Bells Line Rd OBGE	6	6 (4 2)	15	10	2	1	No
7	145	B145 Kooraban St overbridge	3	6 (4 2)	12	6	10	5	No
8	852	T-OBGE01000168: I00 069.984 THI Lawrence H'grave Dr OBGE	10	6 (4 2)	14	7	4	4	No
9*	6298	B6298 Sheahan Bridge	4	7 (3 4)	10	0	15	6	Yes
9*	972	T-OBGE01000566: W00 085.958 LIN GT West Hwy W.Bound OBGE	3	5 (3 2)	15	6	5	5	No
9*	1363	T-OBGE01000214: N00 Pacific Hwy OB Wyong 103.0	6	6 (4 2)	16	8	6	1	No

Sum of all criteria scores shown above contribute to the bridge priority ranking. The higher the criteria score the higher the ranking. Refer to Appendix for attached file containing bridge prioritisation list and analysis.

* denotes bridges that have equal scoring, thus equal rank

6.2

Updated Interim Prioritisation Findings and
Results (February 2026)



Executive Summary

The section presents the ranking of the Top 21 bridges, alongside comparative analysis of all 99 bridges.

Key findings:

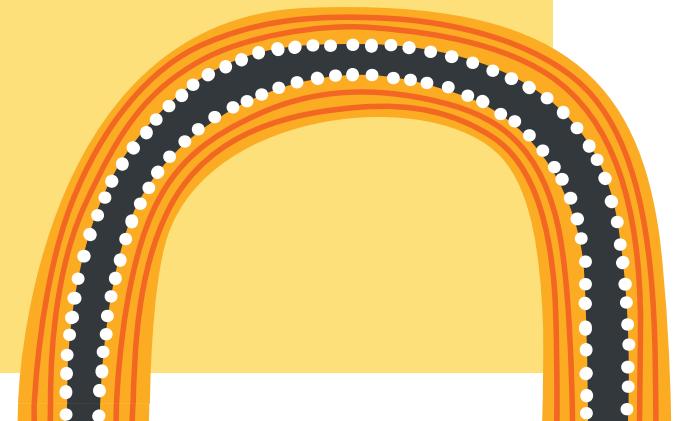
- Bridge rankings remain broadly consistent between the original and updated prioritisation.
- Including multiple user types increases scores for some lower-ranked bridges.
- Missing or inaccurate data may be understating scores for some originally high-ranked bridges.
- While a comprehensive set of user types was applied, not all may be necessary or should be equally weighted, given their differing levels of importance.
- GSRA and RMR ASPs do indicate the prioritisation of bridge investment does use a multi criteria assessment process

Next steps:

- Refine and finalise user types and criteria that add limited value.
- Conduct sensitivity testing of criteria as part of the development of next steps roadmap to ensure there is no prejudicial treatment of data gaps.
- Review the weighting and summation approach.
- Undertake further data gap analysis and targeted data collection, particularly where confidence is low.

6.2.1

Recap: Data Confidence



Findings and limitations: Vulnerability Criteria

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Bridge Health Index (BHI)	Medium	Bridge health criteria and data not common across different modes (road and rail use different measures of asset condition)	Project Team have engaged TfNSW Chief Engineer to determine a common bridge condition measure applicable to all modes
Historical safety incidents	Low	Historical safety incident data was not available for this analysis and was not applied to the interim prioritised list	Project team to identify relevant sources of bridge safety incident data with asset stewards that can be attributed to individual bridges
Remaining lifespan	Medium	Remaining useful life data was not available for the rail bridges in TACP (available for road bridges)	Project team to identify relevant sources of remaining useful data with Country Rail asset steward, including data update for TACP
Heritage status	High	Bridges were identified on the relevant Australian and State Heritage Registers	

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Findings and limitations: Criticality Criteria

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Network role	Medium	Limitations in classifying heavy vehicle classes (only class 1 identified from NHVR). Active transport use and under bridge use determined using Google Maps.	Project team to work with Freight team to source more accurate heavy vehicle classification and identify more efficient / reliable GIS/database options available with TfNSW stakeholders for under bridge use.
Network safety	Low	Network safety data was not available for this analysis and was not applied to the interim prioritised list	Project to identify with TfNSW stakeholders relevant data source for bridge network safety use, including emergency management and evacuation functions.
Public transport routes	Medium	Google Maps and public transport network maps were sourced to identify number of public transport services over each bridge	Project to identify with TfNSW stakeholders more relevant and accurate data source for bridge's contribution to public transport service such as available seat kilometres, school bus routes, and intermodal connectivity.
Traffic density and load	Medium	AADT and BIS data was used to determine traffic volumes and heavy vehicle loads for road bridges. SWTT was used for passenger train volumes, and data was not available for freight train volumes.	Project to identify with TfNSW stakeholders more relevant and accurate data source for heavy vehicle loads (ESAL), passenger/freight rail volume and activity transport traffic for each bridge.
Route alternatives	Medium	Google Maps was sourced to identify and calculate route alternatives to determine total detour time for each bridge for passenger and active transport.	Project to identify with TfNSW stakeholders more relevant and accurate traffic modelling / data source for calculating route alternatives for all transport users.

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Findings and limitations: Criticality Criteria (cont.)

Interim bridge prioritisation has identified limitations and improvement opportunities based on the limited data sources available for the 99 Vulnerable Bridges Sample Set

Recommendations outlined below in response to the findings are considered in the Final Report

Criteria	Data Confidence	Findings / Limitations	Recommendation
Modal inter-dependencies	Low	Modal interdependency data was not available for this analysis and was not applied to the interim prioritised list	Project to identify with TfNSW stakeholders relevant data source to assess how the failure of a bridge may contribute to the mode shift of users and relationship/applicability with the route alternative metric.
Proximity to critical services	Medium	Identification of critical services / infrastructure supported by each bridge was conducted using NSW Open Map Data	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for identification of critical services
Third-party criticality	Medium	Identification of third-party assets on bridges was conducted using Google Maps for each individual bridge	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for identification of third-party assets
Community isolation	Medium	Identification population that is isolated if a bridge fails was conducted using Google Maps and ABS population data for each individual bridge	Project team to explore more efficient and reliable GIS/database options available with TfNSW stakeholders for assess community/population isolation impacts (where no alternative routes are available)

Note: Data confidence is representative of the available data at the time of the interim bridge prioritisation of the 99 Vulnerable Bridges Sample Set. The identified recommendations aim to improve data confidence and results for the final bridge prioritisation application.

Data availability & confidence for each user type and criteria

	Bridge Health Index (BHI)	Historical safety incidents	Remaining lifespan	Heritage status	Network role	Network safety	Public transport routes	Traffic density and load	Route alternatives	Modal interdependencies	Proximity to critical infrastructure	Third-party criticality	Community isolation
Over - Road - Passenger	Medium	Low	Medium	High	High	Low	Medium	Medium	Medium	Low	Medium	Medium	Medium
Over - Road – Heavy vehicles Class 2 & 3					Low	Low	N/A	High	Unavailable	Low	N/A		N/A
Over - Road – Heavy vehicles Class 1					Medium	Low	N/A	Unavailable	Unavailable	Low	N/A		N/A
Over - Active Transport – Pedestrians & cyclists					Medium	Low	N/A	Unavailable	Medium	Low	N/A		N/A
Over - Rail - Passenger trains					Medium	Low	Medium	Medium	Unavailable	Low	N/A		N/A
Over - Rail - Freight trains					Medium	Low	N/A	Unavailable	Unavailable	Low	N/A		N/A
Under - Road - Passenger					Medium	Low	Medium	Unavailable	Medium	Low	Medium		Medium
Under - Road – Heavy vehicles Class 2 & 3					Unavailable	Low	N/A	Unavailable	Unavailable	Low	N/A		N/A
Under - Road – Heavy vehicles Class 1					Low	Low	N/A	Unavailable	Unavailable	Low	N/A		N/A
Under - Active Transport – Pedestrians & cyclists					Medium	Low	N/A	Unavailable	Medium	Low	N/A		N/A
Under - Rail - Passenger trains					Medium	Low	Medium	Medium	Unavailable	Low	N/A		N/A
Under - Rail - Freight trains					Low	Low	N/A	Unavailable	Unavailable	Low	N/A		N/A
Under- Water Transport – Passenger & private vessels					Unavailable	Low	Medium	Unavailable	Medium	Low	N/A		N/A
Under - Water Transport – Freight & commercial vessels					Unavailable	Low	N/A	Unavailable	Medium	Low	N/A		N/A

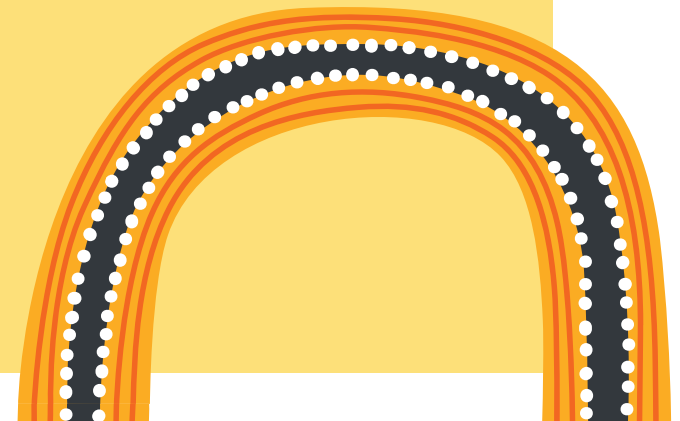
Note: **Third-party criticality** does not vary by user type. While it is a **Criticality** criterion (i.e., it relates to the function of the bridge, not the physical attributes of the bridge), the bridge is evaluated on the whole for this criterion, similar to the four **Vulnerability** criteria.

Community isolation and **Proximity to critical infrastructure** are only evaluated for *Road – Passenger* user types (both Over and Under the bridge); other user types are not relevant for these criteria

There is some variation between Over and Under for user types in **Network role** and **Traffic density and load**, due to varying data availability and confidence

6.2.2

Top 21 bridges ranking
comparisons



Top 21 bridges

The original rankings were based on High, Medium, and Low categories, with 21 bridges classified as High.

Ranking comparisons

Bridge Number (BIS only)	Bridge Name	ORIGINAL RANK (Top 21)	RANK (UPDATED)	VULNERABILITY Subtotal score	CRITICALITY Subtotal score	GRAND TOTAL SCORES
1518	B1518 Forster/Tuncurry Bridge	H	1	7	47	54
1365	B1365 Swansea north bound	H	2	8	43	51
12555	B-12555 - Footbridge connecting Meadowbank and Rhodes will be transferred from Crown Lands to TfNSW	H	2	14	37	51
OBN00177A	New England Highway Overbridge Tarro	H	4	1	49	50
6298	B6298 Sheahan Bridge	H	9	4	41	45
972	T-OBGE01000566: W00 085.958 LIN GT West Hwy W.Bound OBGE	H	9	3	42	45
44	T-OBGE01000809: N30 020.634 TUR Pacific Hwy OBGE	H	9	5	40	45
OBN00167A	Maud Street Overbridge Sandgate	H	13	3	41	44
OBN00192A	Church Street Overbridge Maitland	H	15	5	38	43
3215	B3215 Swan Hill Bridge	H	16	11	29	40
942	Bridge 941 and 942 road over rail bridge in Wollongong leading into the port	H	16	7	33	40
2322	B2322 Old Grafton Bridge	H	18	13	26	39
3244	B3244 Tooleybuc Bridge	H	21	11	27	38
941	Bridge 941 and 942 road over rail bridge in Wollongong leading into the port	H	21	5	33	38
98	B98 Bentley's Bridge	H	25	11	25	36
OBC03166A	Industrial Drive Overbridge Port Waratah	H	31	1	32	33
875	B875 Hampden Bridge, Kangaroo Valley	H	34	11	21	32
4122	B4122 Angus Bridge	H	55	5	22	27
4767	B04767 Boothaguy Creek Bridge	H	60	3	22	25
4660	B4660 Scabbing Flat Bridge	H	69	10	13	23
938	T-OBGE01000938: I10 085.068 CNS Springhill Rd Trilp OBGE	H	73	3	19	22

These bridges were originally in the Top 21 list, but not under the new prioritisation framework.

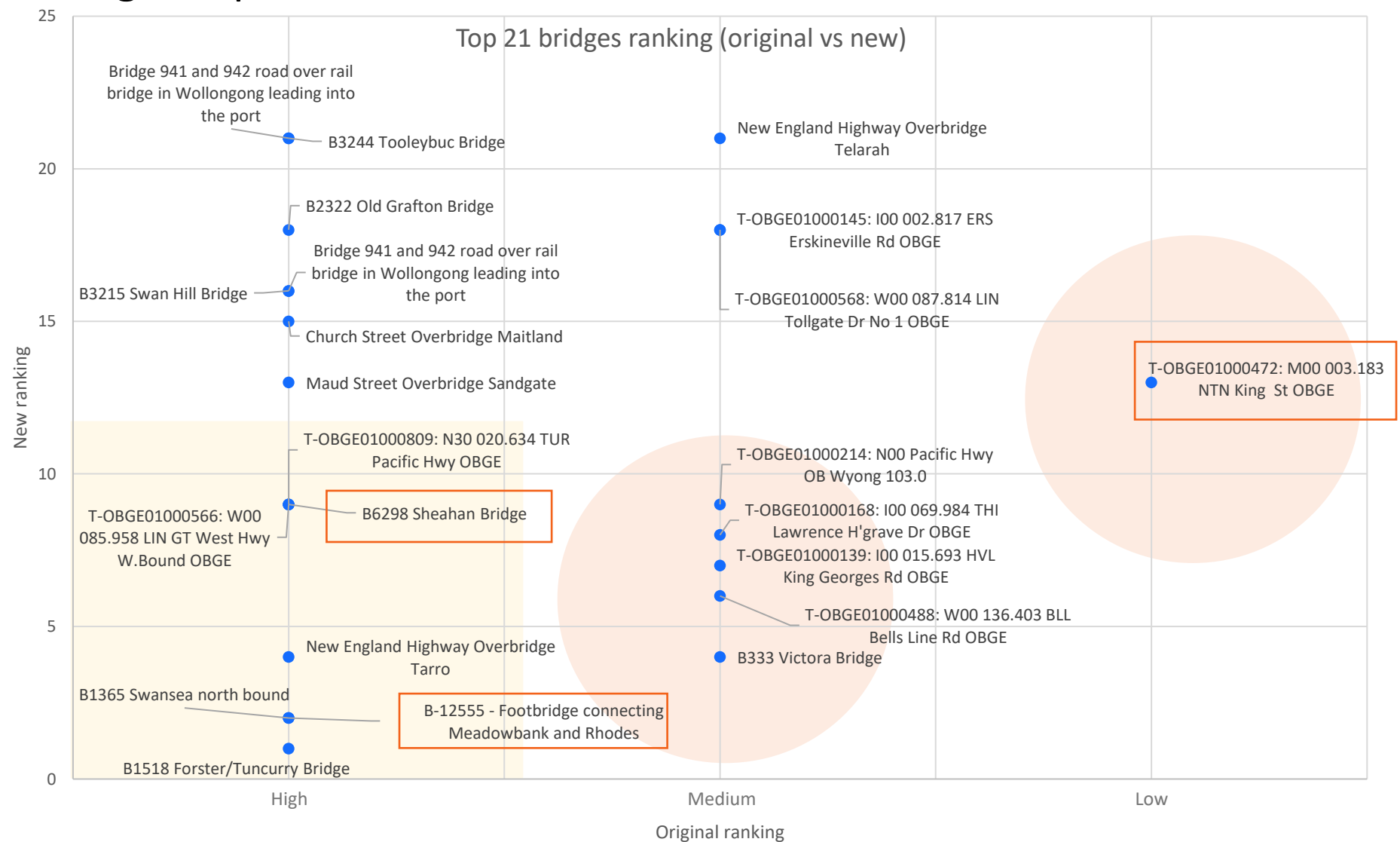
Details are on the next page.

Top 21 - Why the bridges drop out of the Top 21?

Bridge Number (BIS only)	Bridge Name	ORIGINAL RANK (Top 21)	RANK (UPDATED)	VULNERABILITY Subtotal score	CRITICALITY Subtotal score	GRAND TOTAL SCORES	Why drop out of the Top 21?
98	B98 Bentley's Bridge	H	25	11	25	36	- Limited user types. No users under the bridge. Over the bridge: passenger car, HVs (Class 1, 2 and 3), and active transport. Leading to low scores on network role, traffic density and load etc - Quick detour: <20mins detour for passenger cars, leading to low score for Route Alternatives criteria 2 Public bus routes (although school bus routes not considered) - Although medium vulnerability score: BHI=5, poor; Remaining lifespan: 6% life remaining; Heritage listed
OBC03166A	Industrial Drive Overbridge Port Waratah	H	31	1	32	33	Low vulnerability score: BHI data missing; Remaining lifespan: 60% life remaining; Not heritage listed
875	B875 Hampden Bridge, Kangaroo Valley	H	34	11	21	32	Medium criticality score: - Limited user types. No users under the bridge. Over the bridge: only passenger cars and active transport - Although require significant detour time
4122	B4122 Angus Bridge	H	55	5	22	27	Low vulnerability score: - Remaining lifespan: 48% life remaining; Not heritage listed Medium criticality score: - Limited user types. No users under the bridge. Over the bridge: passenger cars and HVs. Leading to low scores in criteria such as traffic density and load, alternative routes 1 Public bus route (although school bus routes not considered)
4767	B04767 Boothaguy Creek Bridge	H	60	3	22	25	Low vulnerability score: - BHI=1, good; Historical safety incident: missing data; Remaining lifespan: 41% life remaining; Not heritage listed Medium criticality score: - Limited user types. No users under the bridge. Over the bridge: passenger cars and HVs. Leading to low scores in criteria such as traffic density and load, alternative routes - Not on public transport routes (although school bus routes not considered)
4660	B4660 Scabbing Flat Bridge	H	69	10	13	23	Medium vulnerability score: poor BHI and low remaining life; but not heritage listed Low criticality score: - Limited user types. No users under the bridge. Over the bridge: passenger cars, HV Class 2&3, active transport. Leading to low network role - Missing data on HV usage. Missing data on traffic density and load. - Not on public transport routes (although school bus routes not considered)
938	T-OBGE01000938: I10 085.068 CNS Springhill Rd TriIp OBGE	H	73	3	19	22	Low vulnerability score: - BHI=1, good; Historical safety incident: missing data; Remaining lifespan: 30% life remaining; Not heritage listed Low criticality score: - Limited user types. Over the bridge: freight trains; Under: passenger cars, HVs and active transport. - Short detour time for passenger cars - Missing data on HV usage under the bridge - Criteria such as proximity to critical infrastructure and third-party criticality not applied – data gap

Top 21 – Most bridges maintain high ranking, noting movements in a few bridges

Ranking comparisons

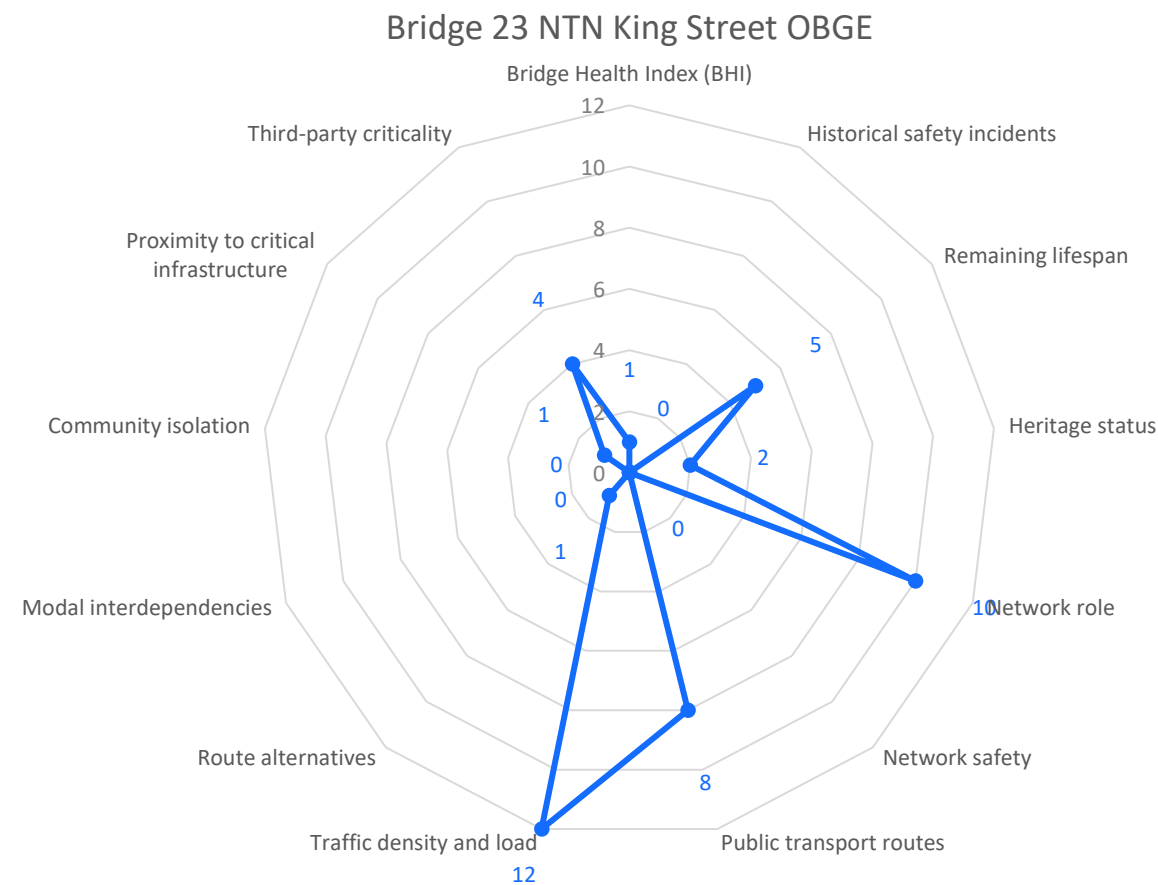


- Most bridges previously ranked in the Top 21 remain high-ranking under the new framework.
- Seven bridges previously rated medium now move into the Top 21.
- One bridge previously rated low also moves into the Top 21.

The following slides highlight selected bridges to illustrate the key drivers behind these ranking changes.

Bridge Number 23 (BIS) – NTN King Street OBGE

Original rank: Low >> Updated rank: 12 out of 99 bridges



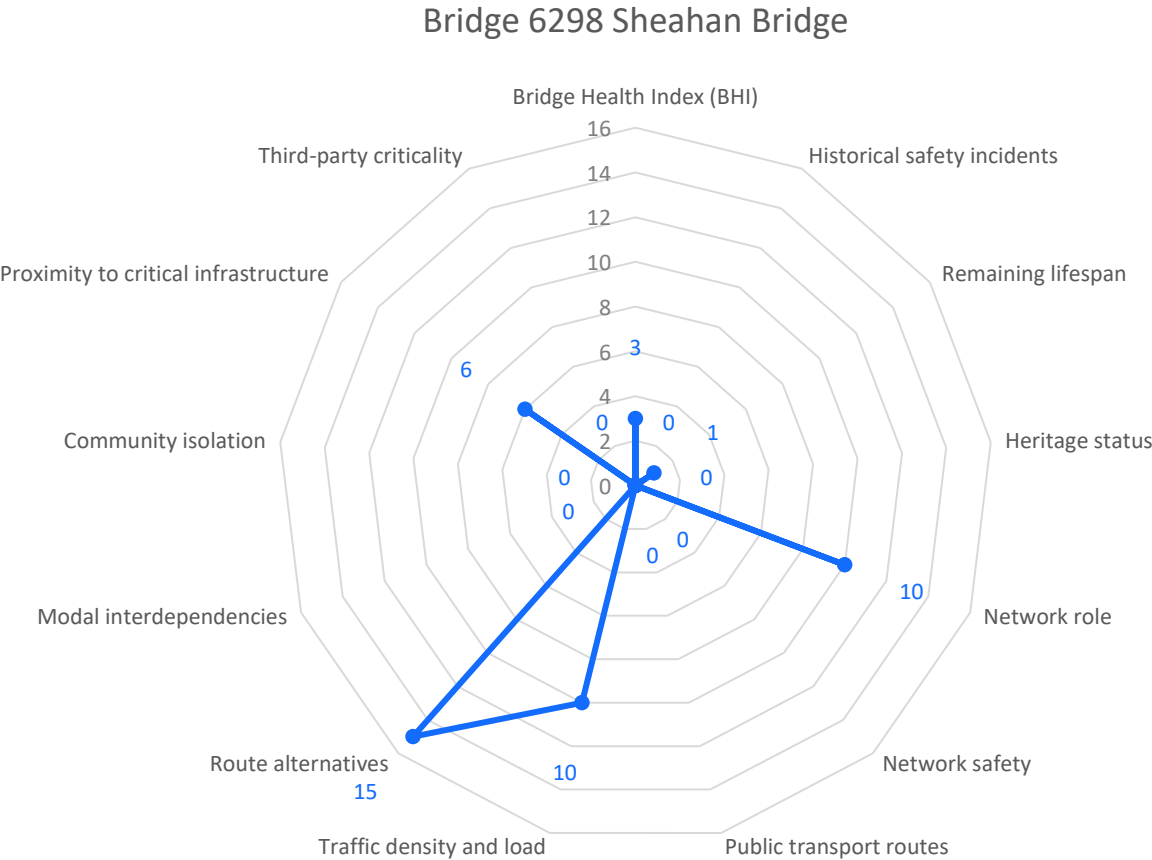
Key factors driving the bridge’s significantly **higher priority under the new prioritisation** framework include:

- **Remaining lifespan:** It receives a full score, with TACP data indicating a –56% remaining life (no evidence of recent renewal).
- **Traffic density and load:** Extremely high scores due to four train lines passing beneath the bridge. Also incorporated score for heavy-vehicle annual tonnage (TransIT data).
- **Network role:** Strong score reflecting multiple user types, including its role on the A36 highway and a major metropolitan rail line.
- **Public transport importance:** High score driven by 10 bus routes using the bridge and passenger rail operating underneath.

Over - Road – Passenger	Over - Road – Heavy vehicles Class 2 & 3	Over - Road - Heavy vehicles Class 1	Over - Active Transport – Pedestrians & cyclists	Over - Rail - Passenger trains	Over - Rail - Freight trains	Under - Road - Passenger	Under - Road – Heavy vehicles Class 2 & 3	Under - Road - Heavy vehicles Class 1	Under - Active Transport – Pedestrians & cyclists	Under - Rail – Passenger trains	Under - Rail – Freight trains	Under- Water Transport - Passenger & private vessels	Under - Water Transport - Freight & commercial vessels
Yes	Yes	No	Yes	No	No	No	No	No	No	Yes	Yes	No	No

Bridge Number 6298 (BIS) – B6298 Sheahan Bridge

Original rank: High -- Updated rank: 9 out of 99 bridges



Sheahan bridge remains in the Top 21 bridges under the updated ranking. Key drivers are:

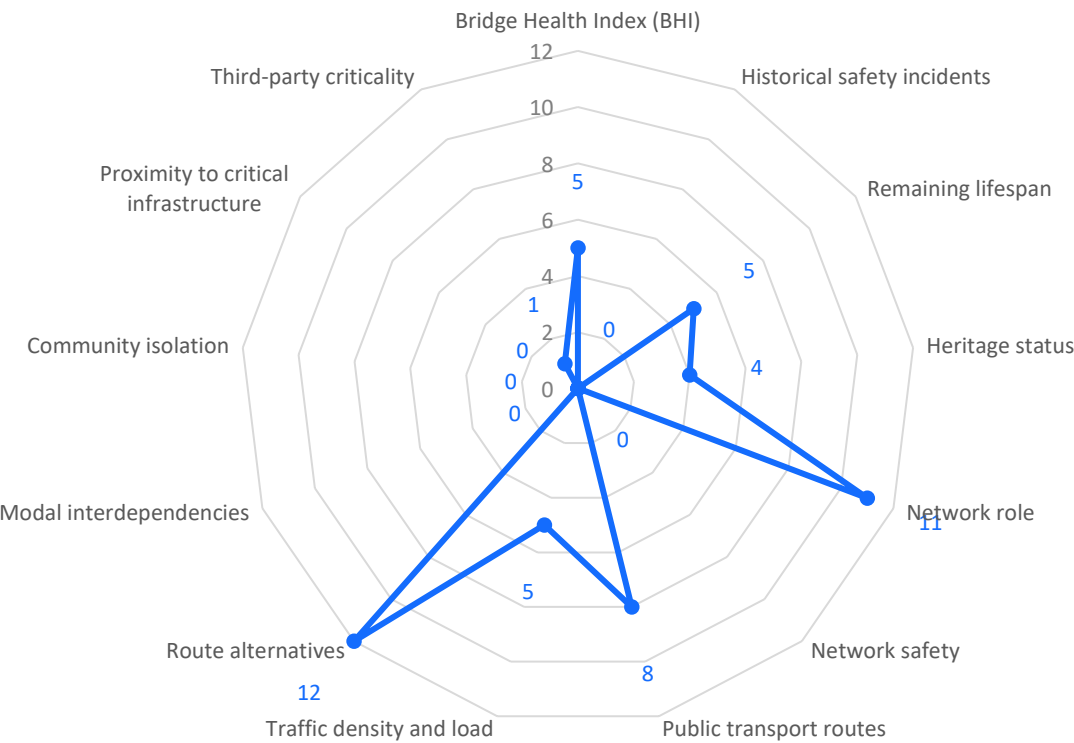
- There are multiple user groups operating both over and under the bridge:
- **Route alternatives** score extremely high, as detours exceed 90 minutes for road users below the bridge.
- Route alternatives score also considered that there are **no viable alternatives for water transport** (passenger services, freight operators, or commercial vessels).
- The bridge supports a **critical network role** as part of the Hume Highway (M31), including approved access for Class 1 heavy vehicles.
- **Traffic volumes and freight loads** are substantial, supported by recorded freight tonnage data (TraNSIT).

Over - Road - Passenger	Over - Road - Heavy vehicles Class 2 & 3	Over - Road - Heavy vehicles Class 1	Over - Active Transport - Pedestrians & cyclists	Over - Rail - Passenger trains	Over - Rail - Freight trains	Under - Road - Passenger	Under - Road - Heavy vehicles Class 2 & 3	Under - Road - Heavy vehicles Class 1	Under - Active Transport - Pedestrians & cyclists	Under - Rail - Passenger trains	Under - Rail - Freight trains	Under- Water Transport - Passenger & private vessels	Under - Water Transport - Freight & commercial vessels
Yes	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	Yes	Yes

Bridge Number 12555 - Footbridge connecting Meadowbank and Rhodes

Original rank: High -- Updated rank: 2 out of 99 bridges

Bridge Number 12555 - Footbridge connecting Meadowbank and Rhodes



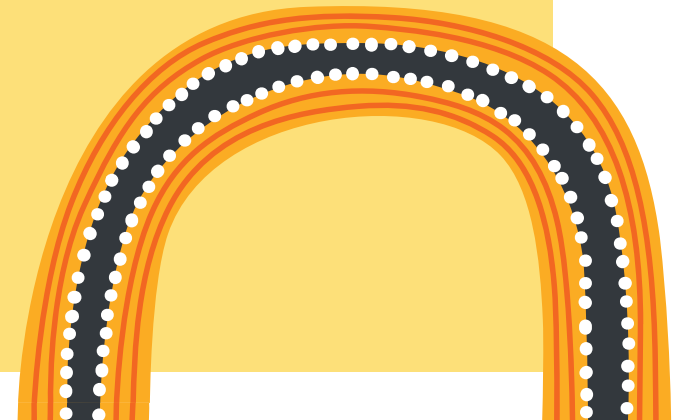
Key drivers of the bridge's high priority ranking include:

- **Route alternatives score** is extremely high due to no alternatives for water transport users and a 25-minute detour for active transport.
- **Network role score** is high: the bridge carries the T9 metropolitan line, an ARTC freight route with ~88 weekly services, and a shared pedestrian path.
- **Remaining life scores** at the maximum level: TACP indicates ~40% remaining life, with uncertainty about recent renewals.
- **BHI** also scores at the maximum level: BHI = 5 (poor).

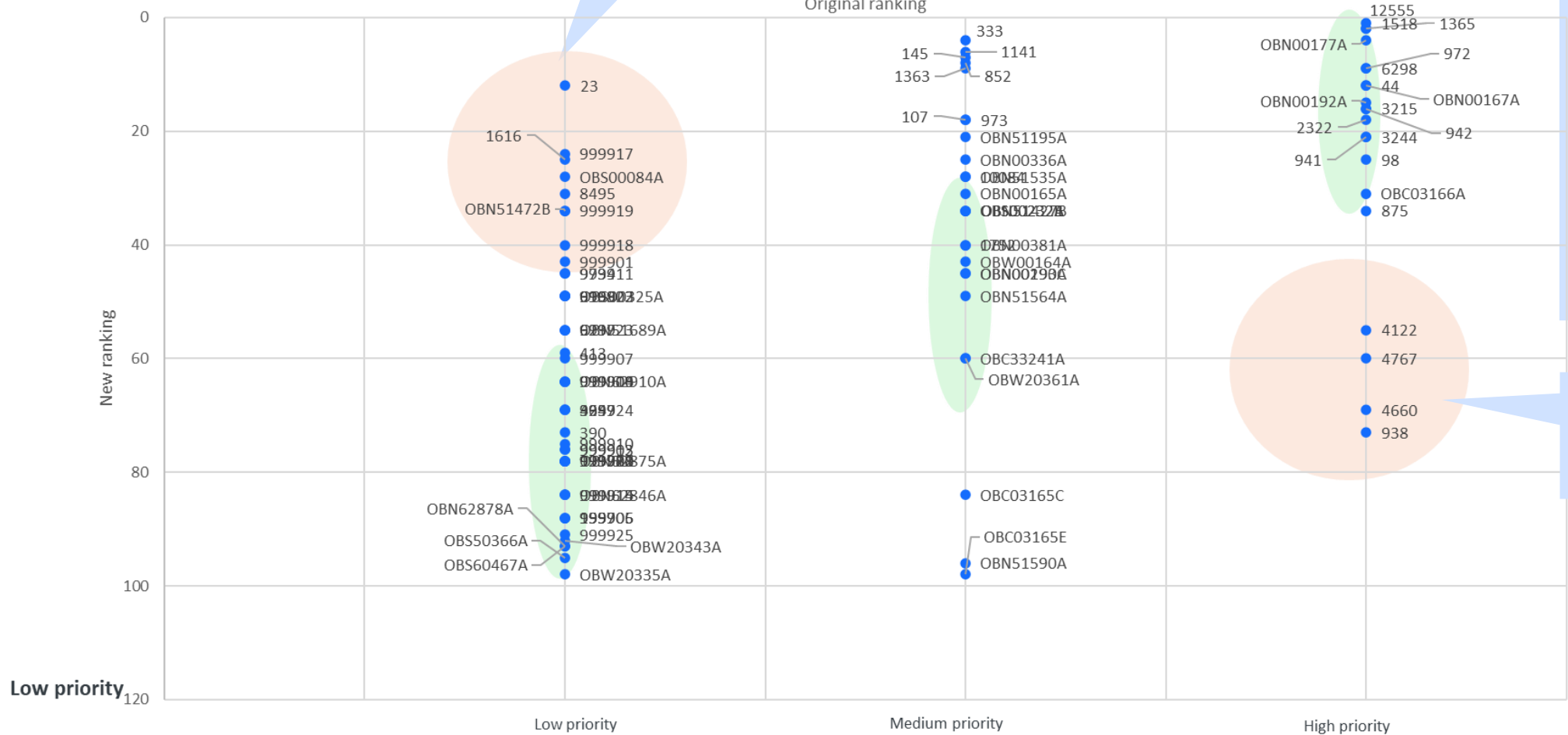
Over - Road - Passenger	Over - Road - Heavy vehicles Class 2 & 3	Over - Road - Heavy vehicles Class 1	Over - Active Transport - Pedestrians & cyclists	Over - Rail - Passenger trains	Over - Rail - Freight trains	Under - Road - Passenger	Under - Road - Heavy vehicles Class 2 & 3	Under - Road - Heavy vehicles Class 1	Under - Active Transport - Pedestrians & cyclists	Under - Rail - Passenger trains	Under - Rail - Freight trains	Under - Water Transport - Passenger & private vessels	Under - Water Transport - Freight & commercial vessels
No	No	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes	Yes

6.2.3

99 bridges ranking comparisons



Ranking comparisons



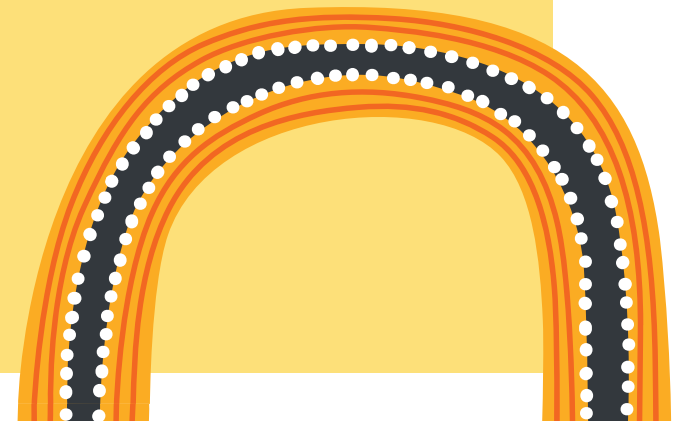
99 bridges ranking (original vs new)

- Clear clustering under the new ranking, broadly consistent with the original high/medium/low groups.
- Wider score spread reflects the expanded criteria, including a broader set of user types and more comprehensive vulnerability and criticality measures.
- Summing scores across multiple user types and criteria naturally produces a wider overall score range. This can be adjusted in the weighting process.

- Clusters of bridges aligning with previous rankings
- Bridges with major shifts in ranking

6.2.4

ASP & TACP data assessment summary



GSRA ASP

The GSRA ASP clearly identifies the number of bridge assets with the portfolio, categorised and profiled by zone location, age and assessed condition. The condition ratings align with current TfNSW standards (1-5). The few assets in a poor or worse condition identified with commentary as to the current actions being undertaken.

The GSRA works program is informed through the use of a multi-criteria assessment process to identify and prioritise a works program for Bridges Assets in their portfolio domain.

The GSRA Rough Cut Capacity Plan utilises bridge information is sourced from the BIS and supported with a Risk Assessment (in line with TERM categories). Additional criteria and detail is built into the methodology include Statutory obligations (inc heritage), policy objectives or outcome, sustainability, performance, innovation and road use category. Values are applied to the criteria and a resulting weighted score is achieved. This identification process provides context for future projects. The road use or travel level category does include a level of customer impact in the calculation methodology but it is not directly quantified.

Project funding (OPEX or CAPEX) for bridge assets is proposed with several Key Result Areas being achieved by the project. These include Performance, Safety, Sustainability and Customer Focus. Projects and scenarios are forecast over the short to medium term, up to ten years.

RMR ASP

The RMR ASP clearly identifies the number of bridge assets with the portfolio, categorised and profiled by zone location, age, assessed condition and construction material. The condition ratings align with current TfNSW standards (1-5). Asset Management Objectives are mapped to Outcome KPI's. Bridge Health Index is a KPI were 99% of Bridges across the portfolio are required to be in a "Fair" condition or better.

The RMR overall works program, including bridges, is informed by utilising the Enterprise Outcomes-based prioritisation principles to guide the investment approach.

The RMR ASP details that they use a **Risk and Condition enabled approach across the portfolio**. The Asset Strategy and Maintenance & Delivery teams work collaboratively to complete analysis of observed and measured condition in determining the appropriate rating and also predict the future condition based on the forecast use or demand. and impact in the calculation methodology but it is not directly quantified.

Asset Criticality for bridges is based on the function (road travel speed and strategic importance) and traffic volume (inc percentage of heavy vehicles) and a sub-network ranking can be derived and utilised. RMR utilise a Road Network Classification to rank the criticality

Project funding (OPEX or CAPEX) for bridge assets is proposed with several Key Result Areas being achieved by the project. These include Performance, Safety, Sustainability and Customer Focus. Projects and scenarios are forecast over the short to medium term, up to ten years.

Sydney Trains or Country Regional Network (Rail) ASPs not available for comparison

TACP Assessment

TACP identifies 11,850 bridges that TfNSW organisations are responsible for. The information in the TACP databases has come from the numerous sources. Data gaps in TACP indicate missing data from the external original collated source information.

Road and Rail bridges, of various types (steel, timber, Brick etc) make up the majority of the portfolio across the state. Other bridges in the portfolio include: pedestrian foot bridges and subway underpasses, stock underpasses and other fauna passageways, bridges that support services (water, telco and power) and service crossing on industrial lands.

For the portfolio of 11,850 bridges, these are some of the key criteria used in the prioritisation model:

94% have the **Asset Owner** identified and 6% classified as “Others”

100% have the **Asset Custodian** Identified

42% have the **Primary Maintainer** identified, **58%** are **associated with Road Network Resilience**

74% have the specific **Location** fields complete, but associated information including GIS coordinates, location corridor and Asset Description including a location minimises location error gap.

88% have **GIS coordinates** completed.

22% have a **Condition** identified as “Null”. This information will need to be updated across the portfolio on a 6-12 monthly basis.

Multiple maintainers have well embedded, approved processes and methodologies for assessing condition of these assets. **The reported description of condition is inconsistent.** Some maintainers use a Bridge Health Index (BHI) other do not.

42% have **Asset Criticality identified as “Null”**. Criticality is described and used by different operators and maintainers. It can involve multiple complex references including a road traffic ranking, road class description, number of timetabled services across a location. Some quantified thresholds are described in the Security of Critical Infrastructure Act (2018) legislation.

Other observations from TACP data set

No current information exists (**0%**) for the date of **Last Condition Assessment** in the TACP data Set. This information will need to be updated across the portfolio on a 6-12 monthly basis.

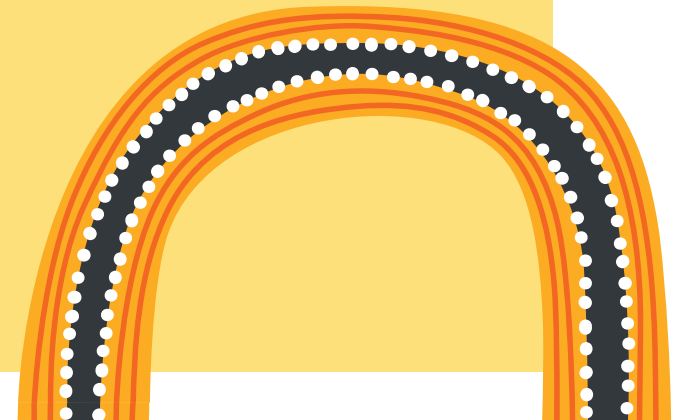
Less than 70% of the Bridge portfolio have **Construction or Commission Dates** and Service life End Date were not included in the evaluated data set, so an assessment of data effectiveness is incomplete, although these categories do exist in the TACP data Dictionary. As a Bridge may be beyond its original design life, upgrades, renewals and appropriate maintenance can will extend operating life.

Complex locations are asset locations that can have multiple transport types at the same site. Bridges can have road, rail, pedestrian surface or subway paths etc structural elements. Each asset type is recorded separately not indicating a parent child relationship.

Many of the customer use source inputs into the Bridge Prioritisation Framework do not currently reside in the TACP database

6.2.5

PRELIMINARY DRAFT HURDLE
RATES



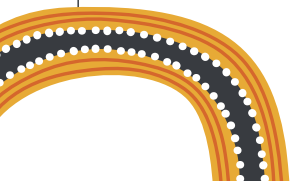
Multi criteria analysis

Hurdle rates and data sources

The hurdle rates and data sources proposed on the following slides are those drafted for preliminary discussions and stakeholder feedback. They have been included here for completeness; they require further refinement, or in the case of the interim prioritisation output have already been amended or refined for applicability.

These indicative hurdle rates were drafted to demonstrate how the MCA criteria *could* be quantified and should be regarded as contextual rather than prescriptive.

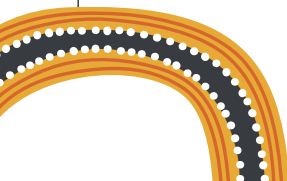
The provided hurdle thresholds or underpinning data sets suggested may not be available now or into the future, nor necessarily the most appropriate for the respective criterion / user type; however, they may nonetheless be valuable for informing future data uplift work and subsequent integration with the prioritisation framework.



Criticality

DRAFT HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Network role <i>Importance of the bridge in the broader transport network. Reflects the role of the bridge in maintaining national, state and local transport continuity. Includes if the bridge forms part of the National Land Transport Network, a critical freight or passenger corridor, or a Strategic Cycleway Corridor.</i>	Road – passenger	Part of National Land Transport Network	- On a state road or - On critical passenger corridor	On a regional road	N/A	On a local road	Local access only	- BIS data “BRIDGE_RD – CLASS_DESC” - National Land Transport Network
	Road – class 2 and 3 HV	Part of National Land Transport Network	On Strategic Freight Network or key freight route	On Regional Freight Corridor	N/A	Provides local freight access to one precinct	Limited or seasonal freight use	Heavy Vehicle Access Policy
	Road – class 1 HV and hazardous freight	On Approved OSOM or Dangerous Goods Route	Alternative or designated OSOM detour route	Connects OSOM-approved roads but not core	N/A	Infrequent OSOM passage	Not suitable / restricted route	- Heavy Vehicle Access Policy - ROAD RULES 2014 - REG 300-2 NSW rule: carriage of dangerous goods in prohibited areas
	Rail – passenger trains	On a major interstate or metropolitan rail line	On a regional main line carrying passenger services	On a secondary passenger line with moderate service frequency (every 30 mins)	On a branch line with low-frequency (every hour)	Seasonal operations or every more than 1 hour	Disused or abandoned track	Pending data: TACP “Transport Network Code”
	Rail – freight trains	On Interstate or Core Freight Rail Line	On a regional main line carrying freight services	On a freight-only regional line	On a branch line with low-frequency	Occasional freight only	Disused or abandoned track	- ARTC, CRN data - National Network for Interoperability
	Active transport – pedestrians and cyclists	State-wide or national active transport corridor, critical for network continuity. Vital link for inter-city/regional travel	Key link in Strategic Cycleway Corridor or major multi-modal corridor	Dedicated cycleway	Shared cycle path/pedestrian walkway	No separated cycleway Painted cycle lane only, or no cycle lane Standard pedestrian footpath	No pedestrian/cycle access; not part of any walking/cycling network	TfNSW mobile active transport demand data
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							



Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Network safety <i>The degree to which the bridge supports network safety outcomes, including emergency management and evacuation functions. Considers contribution of the bridge to safety by design hierarchy (e.g., avoid level crossings, grade or modal separation).</i>	Road – passenger	- Bridge underpins state or regional network safety; or - Only viable evacuation or emergency route for > 5,000 people; or - Failure would cause severe safety and resilience consequences.	- Bridge is critical for maintaining safe operation of a corridor; or - Loss would isolate > 1,000 people; or - Integral to regional emergency response.	- Directly improves safety outcomes (e.g., replaces level crossing, removes high-risk conflict point); or - Serves local emergency; or - evacuation route for > 500 residents or key facility (e.g., hospital, school).	- Some contribution to safer network operations (e.g., improves sightlines or access); or - Partial role in local emergency access.	- Minimal safety relevance; or - Alternate safe crossing exists; or - Not part of emergency or evacuation network.	- No identifiable network safety function; or - Bridge failure or absence has negligible safety impact (e.g., < 100 daily users).	ABS population data
	Road – class 2 and 3 HV	- Essential freight corridor; or - Loss isolate major logistics hub	Critical link for freight safety continuity or emergency supply routes	Improves freight safety (e.g., grade separation, removes tight curve/low clearance)	Minor improvement in freight route safety or redundancy	Alternate compliant freight route nearby	No freight safety relevance; not on a designated freight route	
	Road – class 1 HV and hazardous freight	Only viable OSOM or Dangerous Goods (DG) route within 50 km; loss prevents compliant movement	Primary OSOM/DG redundancy route	Supports safe OSOM detours or regional emergency bypass	Partial OSOM relevance (permit use)	Limited OSOM role; alternative nearby	No identifiable network safety function	
	Rail – passenger trains	Eliminates high-risk interface (e.g., replaces level crossing); essential for network safety/resilience during emergencies	Critical to maintaining safe rail corridor; failure isolates major station/line section	Improves operational safety (clearances, flood resilience)	Local safety or evacuation relevance	Alternate safe crossing/route available	No identifiable network safety function	
	Rail – freight trains	Core to safe operation of interstate or major freight line; failure disrupts national supply chain	Core to safe operation of regional freight line	Improves freight safety (grade separation, clearance, structure upgrade)	Minor safety contribution on feeder line	Minimal freight safety relevance	No identifiable network safety function	
	Active transport – pedestrians and cyclists	Only safe crossing of major barrier (arterial, rail, or river) on Strategic Cycleway Corridor; loss severs network continuity (> 500 daily trips)	Primary safe route to key destinations (schools, hospitals, stations); > 200 daily users	Improves safety (removes unsafe crossing or adds grade separation); > 100 daily users	Improves local safety (visibility, minor detour)	- Minimal safety relevance; or - Alternate safe crossing exists; or - Not part of emergency or evacuation network.	No pedestrian/cycle safety function	TfNSW mobile active transport demand data
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							

Criticality

Vulnerability

Network role

Network safety

Public transport routes

Traffic density

Route alternatives

Modal interdependencies

Proximity to critical services

Third-party criticalities

Community isolation

Bridge Health Index (BHI)

Historical safety incidents

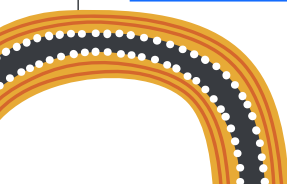
Remaining lifespan

Heritage status

Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Public transport routes <i>The bridge’s contribution to public transport continuity, including available seat kilometres, school bus routes, and intermodal connectivity</i>	Road – passenger	• Bridge carries >10 million ASK/day; or • Supports >5 bus routes including school bus services.	• Bridge carries 5–10 million ASK/day; or • Serves 3–4 bus routes with strong intermodal dependency.	• Bridge carries 1–5 million ASK/day; or • supports 2 regular bus routes with moderate passenger volume.	• Bridge carries 0.5–1 million ASK/day; or • Provides access for 1 bus route	• Bridge used primarily for diversion; or • Occasional bus service, <0.5 million ASK/day.	Bridge has no public transport function; or not used by any scheduled or school bus routes.	• TfNSW NSW Bus Routes • Standard Timetable • Opal data on patronage
	Road – class 2 and 3 HV	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Road – class 1 HV and hazardous freight	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Rail – passenger trains	• Bridge carries >30 million ASK/day	Bridge carries 20m -29 million ASK/day	• Bridge carries 10m -19 million ASK/day	• Bridge carries 3m -9 million ASK/day	• Bridge carries 100 ASK/day-2 million ASK/day	Less than 100 ASK/day	• Standard Timetable • Opal data on trains patronage
	Rail – freight trains	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Active transport – pedestrians and cyclists	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							



Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Traffic density and load <i>Volume and loading (damage) of traffic using the bridge.</i>	Road – passenger	Over 50,000 AADT	40,000-49,999 AADT	30,000-39,999 AADT	5,000-29,999 AADT	Less than 5,000	Nil	- TfNSW's Traffic Volume Viewer - Telematics
	Road – Class 2 & 3 HVs	>10,000 ESALs/day	5,000–10,000 ESALs/day	1,000–5,000 ESALs/day	100–1,000 ESALs/day	<100 ESALs/day	0 ESALs/day	- Heavy Vehicle Access Policy (HVAP) Appendix A - Telematics and permit datasets (e.g., TCA, NHVR) for estimating heavy-vehicle ESAL equivalents
	Road – Class 1 HVs and Hazardous Freight	>5,000 ESALs/day (incl. hazardous freight share > 10%)	2,000–5,000 ESALs/day (hazardous freight 5–10%)	500–2,000 ESALs/day (hazardous freight < 5%)	100–500 ESALs/day	<100 ESALs/day	0 ESALs/day	- Heavy Vehicle Access Policy (HVAP) Appendix A - Telematics and permit datasets (e.g., TCA, NHVR) for estimating heavy-vehicle ESAL equivalents
	Rail – passenger trains	> 80 services/day (intercity/metropolitan main line)	60-79 services/day	30-59 services/day	10-29 services/day	Less than 10 services/day	Nil	Timetables
	Rail – freight trains	>30 million GTK/year, >20 train movements/day	15-30 million GTK/year, 10–20 trains/day,	5-15 million GTK/year, 5–10 trains/day	1-5 million GTK/year, 1-5 trains/day	<1 million GTK/year	No regular train movements; disused or active-transport conversion.	- ARTC or TfNSW Rail Performance Data - Working Timetable
	Active transport – pedestrians and cyclists	> 500 trips/day or on Strategic Cycleway Corridor > More than 1000 pedestrians per day	> 400-499 trips/day or on Strategic Cycleway Corridor > 500-999 pedestrians per day	> 300-399 trips/day or on Strategic Cycleway Corridor > 300-499 pedestrians per day	> 200-299 trips/day or on Strategic Cycleway Corridor > 100-299 pedestrians per day	> Less than 199 trips/day or on Strategic Cycleway Corridor > Less than 99 pedestrians per day	Nil	NSW Strategic Cycleway Corridor
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							

Criticality

Vulnerability

Network role

Network safety

Public transport routes

Traffic density and load

Route alternatives

Modal interdependencies

Proximity to critical services

Third-party criticalities

Community isolation

Bridge Health Index (BHI)

Historical safety incidents

Remaining lifespan

Heritage status

Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Route alternatives Availability of viable alternative routes or crossings if the bridge becomes unavailable, and the delay imposed (detour time).	Road – passenger	- No alternative; or - Detour time >180 min;	Detour time 60-180min	Detour time 30-60min	Detour time 15-30min	Detour time 5-15min	<5 min delay; several equivalent crossings, no material disruption.	Open Street Map
	Road – class 2 and 3 HV	- No alternative; or - Detour time >180 min;	Detour time 60-180min	Detour time 30-60min	Detour time 15-30min	Detour time 5-15min	<5 min delay; several equivalent crossings, no material disruption.	Open Street Map
	Road – class 1 HV and hazardous freight	- No alternative; or - Detour time >180 min;	Detour time 60-180min	Detour time 30-60min	Detour time 15-30min	Detour time 5-15min	<5 min delay; several equivalent crossings, no material disruption.	NHVR National Network Map
	Rail – passenger trains	- No alternative; or - Detour time >180 min; or - Replacement bus service > 12 months;	- Detour time 60-180min; or - Replacement bus service 6-12 months;	- Detour time 30-60min; or - Replacement bus service 3-6 months	- Detour time 15-30min; or - Replacement bus service 1-3 months	- Detour time 5-15min; or - Replacement bus service <1 months	<5 min delay; several equivalent crossings, no material disruption.	NHVR National Network Map
	Rail – freight trains	- No alternative; or - Detour time >270 min;	Detour time 270-180min	Detour time 60-180min	Detour time 30-60min	Detour time 15-30min	<15 min delay; several equivalent crossings, no material disruption.	
	Active transport – pedestrians and cyclists	- No alternative; or - Detour time >90 min;	Detour time 45-90min	Detour time 30-45min	Detour time 15-25min	Detour time 5-15min	<5 min delay; several equivalent crossings, no material disruption.	
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							

Criticality

Vulnerability

Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Modal Interdependencies <i>Assesses how the performance or failure of a bridge may contribute to the mode shift of users. Modal shift will result in broader network impacts, including reliability or delay in the short-term, and increased asset degradation rates in the long-term.</i>	Road – passenger	Prolonged access impediments (>6mo), but no alternate mode available	Prolonged access impediments (>6mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 50%	Short-term access impediments (<6mo) results in alternate mode uptake of >10%	No mode shift	
	Road – class 2 and 3 HV	Prolonged access impediments (>3mo), but no alternate mode available	Prolonged access impediments (>3mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 50%	Short-term access impediments (<1mo) results in alternate mode uptake of >10%	No mode shift	
	Road – class 1 HV and hazardous freight	Prolonged access impediments (>3mo), but no alternate mode available	Prolonged access impediments (>3mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 50%	Short-term access impediments (<1mo) results in alternate mode uptake of >10%	No mode shift	
	Rail – passenger trains	Prolonged access impediments (>6mo), but no alternate mode available	Prolonged access impediments (>6mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 50%	Short-term access impediments (<6mo) results in alternate mode uptake of >10%	No mode shift	
	Rail – freight trains	Prolonged access impediments (>3mo), but no alternate mode available	Prolonged access impediments (>3mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>3mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<1mo) results in alternate mode uptake of 50%	Short-term access impediments (<1mo) results in alternate mode uptake of >10%	No mode shift	
	Active transport – pedestrians and cyclists	Prolonged access impediments (>6mo), but no alternate mode available	Prolonged access impediments (>6mo) results in alternate mode uptake of >80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >50%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 80%	- Prolonged access impediments (>6mo) results in alternate mode uptake of >10%; or - Short-term access impediments (<6mo) results in alternate mode uptake of 50%	Short-term access impediments (<6mo) results in alternate mode uptake of >10%	No mode shift	
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial Vessels							

Criticality

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Heritage status

Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User types	5	4	3	2	1	0	Data source
Proximity to critical services <i>The bridge's role in maintaining access to essential and emergency services, including hospitals, schools. Considers impact on community resilience and continuity of supply during disruptions.</i>	Road – passenger	Only access route to essential services (e.g. hospital, police, school, aged care) within 10 km	One of two access routes to essential services within 10 km	Alternative routes exist but would add >10 km or >15 min travel time	Alternative routes add <10 km or <15 min travel time	Bridge serves non-critical local access only	No link to critical services	Telematics Geospatial estimate
	Road – class 2 and 3 HV	Primary access for HVs to regional facility	Secondary access but provides faster or safer route	Alternate routes available with moderate detour (>10 km or 15–20 min)	Alternate routes easily available (<10 km)	Minimal relevance to emergency or freight services	Not used for emergency or freight access	TfNSW rail network GIS layers
	Road – class 1 HV and hazardous freight	Primary access for emergency services or hazardous freight to regional facility	Secondary access but provides faster or safer route	Alternate routes available with moderate detour (>10 km or 15–20 min)	Alternate routes easily available (<10 km)	Minimal relevance to emergency or freight services	Not used for emergency or freight access	- NSW Rural Fire Service, Ambulance NSW, Police station mapping - TfNSW Freight Network spatial data
	Rail – passenger trains	Only or main rail link to regional health/education hub	Secondary connection supporting essential access	Supports access to employment and education nodes	Minor local rail link	Low relevance to essential access	Not applicable	- TfNSW rail network GIS layers ABS SA2 population/service catchment data
	Rail – freight trains	Key supply chain route for essential goods (e.g. fuel, food, medical supplies)	Secondary freight route serving essential industries	Supports general freight but not essential goods	Minor or local freight line	Non-critical freight line	Not in use	- TfNSW Freight Network NHVR freight route data
	Active transport – pedestrians and cyclists	Only safe walking/cycling access to school, health, or community centre within 5 km	Primary active-transport route linking to services	Alternative but significantly longer or unsafe route (>2 km detour)	Alternative safe route available	Recreational access only	Not used for service access	TfNSW mobile active transport demand data
	Water transport – commuter ferries and private vessels	[hurdle rates will be developed in the following work phase]						
	Water transport – freight shipping and other commercial vessels							

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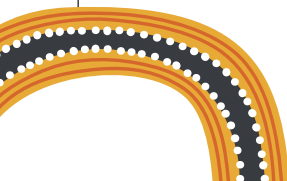
Remaining lifespan

Heritage status

Criticality

DRAFT CRITERIA AND HURDLE RATES

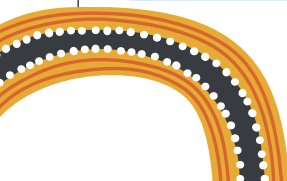
Criteria	User types	5	4	3	2	1	0	Data source
Third-party criticality Quantum and importance of third-party services (e.g., utilities, pipelines, communication assets) or other structures (e.g., carparks, runways) that the bridge supports, where bridge failure would have broader system impact	Applicable to all bridges	Failure would cause regional service outage or require major emergency response.	Bridge supports one critical trunk utility or several secondary services (e.g., primary water/gas mains, or NBN fibre) where disruption affects multiple LGAs or essential services (hospitals, transport control systems, etc.)	Bridge carries local distribution services (e.g., low-voltage power, sewer rising main, local fibre or gas reticulation) affecting a LGA or industrial estate if lost	Bridge supports minor private or service connections, or low-criticality communications cabling with minimal community impact	Bridge includes redundant, disconnected, or easily relocated utility crossings with negligible consequence.	No third-party utilities, pipelines, or communication assets present.	



Criticality

DRAFT CRITERIA AND HURDLE RATES

Criteria	User type	5	4	3	2	1	0	Data source
Community isolation <i>The population that is isolated if a bridge fails, beyond just route alternatives.</i>	Road – passenger	Population impacted >5,000	Population impacted 2,500-4,999	Population impacted 1,000 – 2,499	Population impacted 250 – 999	Population impacted <250	Nil	ABS population
	Road – class 2 and 3 HV	NA	NA	NA	NA	NA	NA	NA
	Road – class 1 HV and hazardous freight	NA	NA	NA	NA	NA	NA	NA
	Rail – passenger trains	NA	NA	NA	NA	NA	NA	NA
	Rail – freight trains	NA	NA	NA	NA	NA	NA	NA
	Active transport – pedestrians and cyclists	NA	NA	NA	NA	NA	NA	NA
	Water transport – commuter ferries and private vessels	NA	NA	NA	NA	NA	NA	NA
	Water transport – freight shipping and other commercial vessels	NA	NA	NA	NA	NA	NA	NA



Criticality

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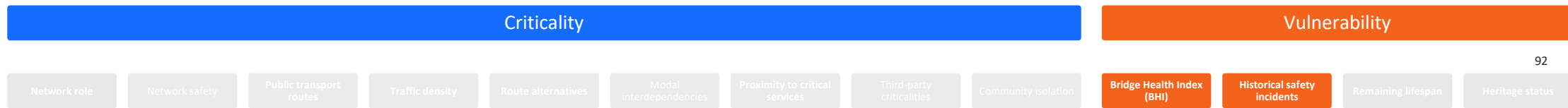
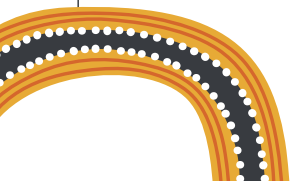
Heritage status

Vulnerability

Proposed hurdle rates and data source

Criteria	5	4	3	2	1	0	Data source
Bridge Health Index (BHI) <i>A broad qualitative indicator of potential risk of structural failure derived from the condition ratings of the bridge elements from the latest inspection of a bridge</i>	Poor	N/A	Fair	N/A	Good	As Built	- BIS data (currently only road bridges information available) "BHI_DESCR" - Comparison of Sydney Trains and TfNSW condition measures
Historical safety incidents* <i>Count of recorded recent incidents or failures; or duration of closure caused by bridge failures</i>	<i>Failure events or emergency Closures occurred: Within the past year; or if the closure lasted more than one day</i>	In the past 1-2 years; or if the closure lasted more than 5 hours	In the past 3-5 years; or if the closure lasted more than 3 hours	In the past 6-8 years; or if the closure lasted more than 1 hours	In the past 8-10 years; or if the closure lasted more than 30 minutes	In the past >10 years or never	TfNSW project team

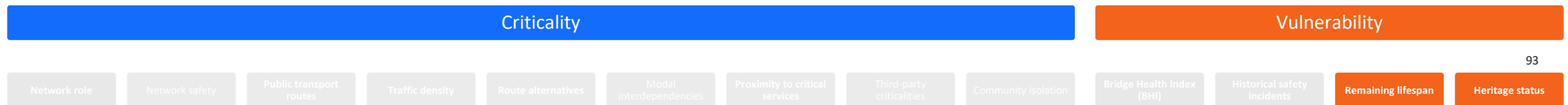
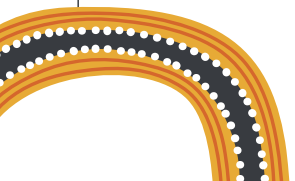
***Note:** This criterion has since been renamed to **Safety incidents and restrictions** and its definition amended. Hurdle rates should include reference to any current access restrictions in place on the bridge, as well as the severity, frequency and outcome (including duration) of recorded or anticipated safety incidents on the bridge's access permissions.



Vulnerability

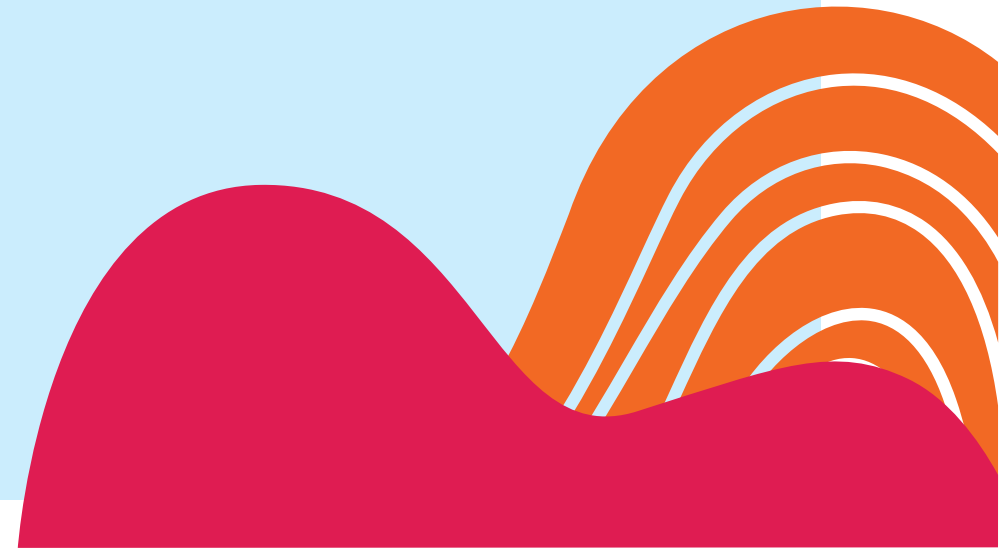
Proposed hurdle rates and data source

Criteria	5	4	3	2	1	0	Data source
Remaining lifespan <i>Remaining life of the asset relative to the total asset design life, expressed as a proportion, accounting for any major maintenance/replacement activities conducted</i>	<5% <u>or</u> <5yrs remaining in the original design life	5%-15%	15%-30%	30%-50%	50%-75%	>75%	- BIS data "YR_COMPLETED, LIFE_SPAN" - Pending: TACP data design life, disposed date, service life end date, warranty end date (requesting the full data set)
Heritage status <i>Is the bridge heritage listed? At what level is it heritage listed?</i>	Yes – National Heritage bridge	Yes – State Heritage bridge	Yes – Local Heritage bridge	No – but bridge is located in a State Heritage curtilage	N/A	No – bridge is not heritage listed, nor in a heritage curtilage	- Asset and Services Plan Road Maintenance Strategy, 2025-2026 - ROM managed State Heritage Register (SHR) listed bridges; - Asset and Services Plan, Greater Sydney Road Assets, 2025-2026



6.3

Bridge Prioritisation Dashboard



TfNSW Bridge AM Assurance and Prioritisation



Bridge Prioritisation Dashboard with real data points (from TACP)

Region

- ☐ NORTH REGION
- ☐ SOUTH REGION
- ☐ SYDNEY REGION
- ☐ WEST REGION

Owner

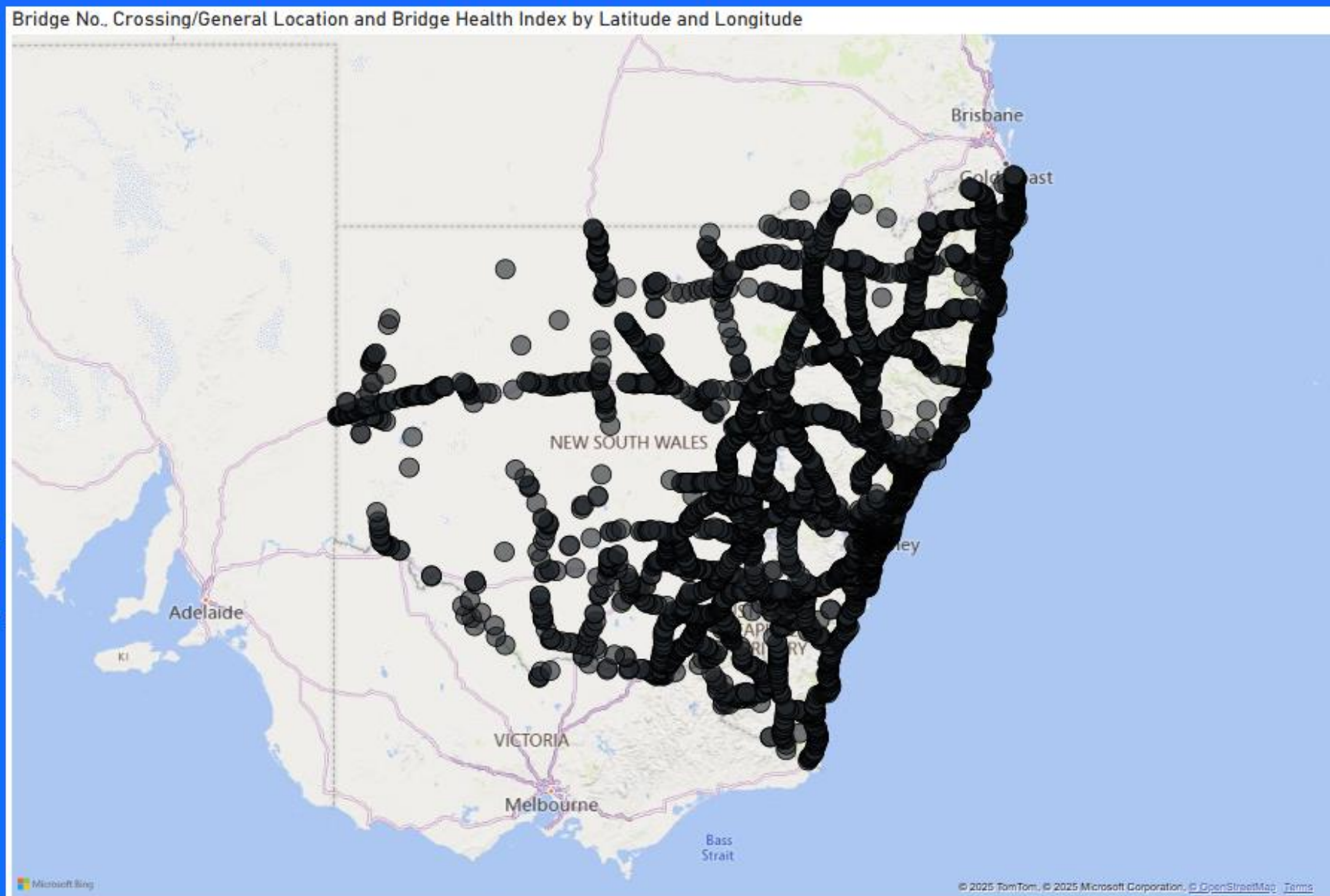
- ☐ A.C.T. Government
- ☐ A.R.T.C
- ☐ Contractor to TfNSW
- ☐ Council (LGA)
- ☐ Country Rail Contracts

Maintainer

- ☐ A.C.T. Government
- ☐ A.R.T.C
- ☐ Contractor to TfNSW
- ☐ Council (LGA)
- ☐ Country Rail Contracts

Bridge Number

Bridge No.	Description
1	ON-LOAD RAMP
2	EB BRIDGE ON HIGH ST
3	WB BRIDGE ON HIGH ST
4	BRIDGE ON MOUNT ST
5	BRDG ON MOUNT ST
6	PED BR AT RIDGE ST
9	BRIDGE ON ERNEST ST
10	BRIDGE ON ERNEST ST
11	BRIDGE ON ERNEST ST



Operational Status

- ☐ Closed To Traffic
- ☐ Completed but Not Opened To Tra
- ☐ Decommissioned (Not controlled ...
- ☐ Opened To Traffic

Material

- ☐ Timber
- ☐ Concrete
- ☐ Steel

Inspection Status

- ☐ Completed
- ☐ Scheduled
- ☐ Overdue

Latest Inspection Date

Start Date vis End Date

Bridge Health Index

- ☐ AS-BUILT
- ☐ FAIR
- ☐ GOOD
- ☐ POOR

Load Type

- ☐ Road
- ☐ Rail
- ☐ Pedestrian

Specialisation Category

- ☐ Heritage
- ☐ Toll
- ☐ Other

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